

MOLTEN SALT REACTOR MICROSCOPIC DELAYED NEUTRON PRECURSOR GROUP
FORMULATION AND EVALUATION

BY

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DISSERTATION

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Abstract

Fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs are delayed neutron precursors (DNPs), which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. DNPs are important for modeling and simulation of MSRs.

Many studies have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. These studies do not account for the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. The effect of neglecting these MSR-specific phenomena is reduced accuracy in the DNP group parameters.

These phenomena are important to consider, as simulations use the DNP group parameters to enable safety, security, and safeguards by design. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for developing an advanced reactor pipeline.

The DNP group parameters for MSRs are resolved using the microscopic, or aggregate or summation, method. This method entails using individual DNP data to reconstruct the DNP groups by summing them together in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half life and are used to generate the longer lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs have a non-negligible impact. Therefore, the longer lived DNP group parameters will vary, while the shorter lived DNP group parameters will remain the same.

A non-linear least-squares solver is used to generate an improved set of DNP group parameters. A specific DNP group parameter set is proposed for a particular set of conditions in the Molten Salt Reactor Experiment (MSRE). Verification is conducted via comparison with total delayed neutron yields and average half-lives from the literature.

The results demonstrate the validity of these MSR phenomena-inclusive DNP group parameters. The methodology used to generate these parameters can be replicated for other MSR conditions, enabling the formation of more accurate sets of DNP group parameters for MSRs. This improves the accuracy of MSR simulations, which is important for safety, security, and safeguards by design as well as for developing an advanced reactor pipeline. Proposed work includes the following: development and evaluation of a more sophisticated model for generating the DNP group parameters, validation via comparison with the MSRE during pump start-up and coast-down, generation of polynomial fits for various parameters used in the DNP group generation, generation of DNP group spectra, uncertainty quantification, analysis of the flux shape effect on DNP group parameters, and evaluation of various datasets on DNP group parameters.

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Chapter 1

Introduction

- Go through history of group parameters (maybe timeline figure?).
- Discuss MSRs.
- "The next step forward for group parameter development is to have group parameters that accurately reflect the physics that occur in fluid-fueled molten salt reactors (MSRs)".
- Introduce the gap [192, 196, 96, 166, 90] that other works do not account for (they investigate movement of DNPs during transient, not for delayed neutron precursor (DNP) group generation).
- Discuss methodology at a high level.
- Discuss results at a high level.
- Discuss conclusions of the work at a high level.

Overview of rest of document

1939 Roberts et al. discovery of delayed neutrons [142]. 1946 initial spectral measurements [30]. In 1947, Hughes et al. initial use of six groups (kinda) at Argonne heavy water pile [73]. In 1957, Keepin used a sample of U235 in Godiva to generate DNP group parameters (improved group fit). Used for a long time [186]. Thus far, only macroscopic approaches have been possible due to insufficient data. For MSRs which have chemical removal of certain elements, tracking individual nuclides is important. In 1989, Brady's thesis used updated data on 271 nuclides identified as DNPs (first attempt at microscopic/summation/aggregate approach). This data was used for ENDF-B/VI was released, and since then there have not been any changes to the DNP group parameters [187, 28]. In 1999, Parish says "Based on the six-group data reported in the literature, it appears that the ENDF/B-VI six-group constants are in need of revision." [123]. In 2014, International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [114]. 2020 Leconte uses more than 400 DNPs with data [95].

Brady's thesis has a good history (up to 1988).

This document should contain tabulated data for various fissile nuclides, neutron irradiation energies, in-core residence times, ex-core residence times, and mixing rates. Additionally, data should be provided on the exact concentrations of each nuclide (up to 99% of the DN yield).

Could also discuss various datasets. [109] - shows various datasets, most important nuclides, and the error associated with them (2012). [76] - Lots of delnu data (2014).

Chapter 2

Background and Motivation

This chapter provides the background and motivation for generation of more accurate DNP group parameters for MSRs and implementation of those parameters for safety, security, and safeguards design purposes. The background information includes information on DNPs and MSR fuel depletion, and the motivation includes safety, security, and safeguards by design. The DNP group parameters in MSRs differ because fuel circulates and undergoes online chemical reprocessing. These effects can be captured in spatially resolved MSR fuel depletion simulations. The existing methodology for DNP group parameter generation does not account for the physical phenomena present in MSR fuel depletion [84, 177, 107, 13, 184, 24, 73, 104, 105, 131, 41, 131, 151, 187, 22, 95]. More accurate DNP group parameters are generated by accounting for these physical phenomena, which is particularly important for safety, security, and safeguards related concerns.

2.1 Delayed Neutron Precursors

Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually. DNPs are an important consideration in nuclear reactors [42]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [10]. In this work, the term DNP will be solely used rather than referring to emitters, as the delay of the neutron emission is effectively only due to the β decay [20, 122]. Classifying between the emitter and the DNP is not relevant for this work.

The delay in neutron emission significantly affects the reactor period and transient behavior. Equation (2.1) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.1)$$

where,

$\dot{n}_{d,i}$ = the delayed neutron activity, or count rate, from DNP i [$n \cdot s^{-1}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from
decay of nuclide i , or the emission probability [-]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of DNP i [*atoms*].

In MSRs, the movement of the fuel salt necessitates spatial resolution for tracking DNP distributions and concentrations. In a solid fuel reactor, such as a light water reactor (LWR), spatial resolution is not necessary to properly model the DNPs, as the fuel remains stationary. Modeling space and time together largely increases computational cost, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity worth, and reactor controllability [42, 43, 91]. Section 2.1.3 shows how the power response and reactivity worth depend on the DNPs. The reactor controllability varies based on the available number of DNPs. If there are fewer delayed neutrons available due to movement of the fuel salt, such as in a MSR, then the reactor period becomes shorter and makes the reactor harder to control.

2.1.1 Measurement

Macroscopic and microscopic approaches are used for measuring DNPs [123, 151]. The macroscopic approach, discussed in detail in Section 2.1.1, uses a fissile sample irradiation to get the delayed neutron count rate, which can be approximated using groups. Conversely, the microscopic approach, discussed in detail in Section 2.1.1, uses the data from many experiments combined together to simulate a fissile sample irradiation and generate the delayed neutron count rate. The microscopic approach is more challenging, as it depends on the work of many experimentalists. However, it offers more detail on the delayed neutron count rate.

Macroscopic Measurement of DNPs

The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. This delayed neutron count rate from this sample can then be measured and used to formulate DNP group parameters. Analysis and collection of macroscopic experiment data is common in the literature [84, 177, 107, 13, 184, 24, 73, 104, 105, 39].

Small samples are used to minimize neutron multiplication within the sample to avoid spectral effects [84]. The spectral effects are due to the differing neutron energy spectra between each DNP group and the incident irradiating neutrons [84]. Another reason for using a small sample is because of physical challenges associated with the rabbit system [29].

Two different types of irradiation are used: pulse and infinite. Pulse irradiation has an irradiation time shorter than the shortest half-life, and it captures the shorter lived DNPs with more accuracy than the longer lived DNPs [24]. Infinite irradiation has an irradiation time longer than several times the longest half-life, and it captures the longer lived DNPs with more accuracy than the shorter lived DNPs [24]. After irradiation, the delayed neutrons from the sample are counted as a function of time. The delayed neutron count rate as a function of time after irradiation, $n_d(t)$, is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where,

$$\begin{aligned} \dot{n}_d(t) &= \text{the delayed neutron count rate } [n \cdot s^{-1}] \\ I &= \text{the total number of DNPs [-]} \\ \epsilon &= \text{the detector efficiency [-].} \end{aligned}$$

Once this delayed neutron count rate is generated, the data is used to generate DNP group and kinetics parameters for that fissile nuclide and neutron energy. Three different approaches have been used to generate DNP group parameters from macroscopic data: exponential stripping (or peeling) [29, 83], least-squares [24, 84], and using half-lives of specific DNPs [157]. Some approaches have even used a combination, such as least-squares fitting with specific half-lives [169]. The exponential stripping, or peeling, method is essentially a more subjective form of least-squares fitting, where exponential curves are subtracted away from the raw data from right to left [135]. The use of specific half-lives to form the DNP groups is less subjective and offers a more physical interpretation of the groups. The least-squares approaches are the most objective approach and generate groups which physically do not embody a single DNP. Instead, they group together the DNPs and effectively capture the overall physical effect, minimizing the objective function. For this work, the least-squares approach is used, as it offers an objective formulation that can be generically applied. The specifics of the least-squares approach used to form the DNP parameters is discussed in Section 2.1.2. Figure 2.1 (recreated from [84]) shows an experimental setup for collecting delayed neutrons.

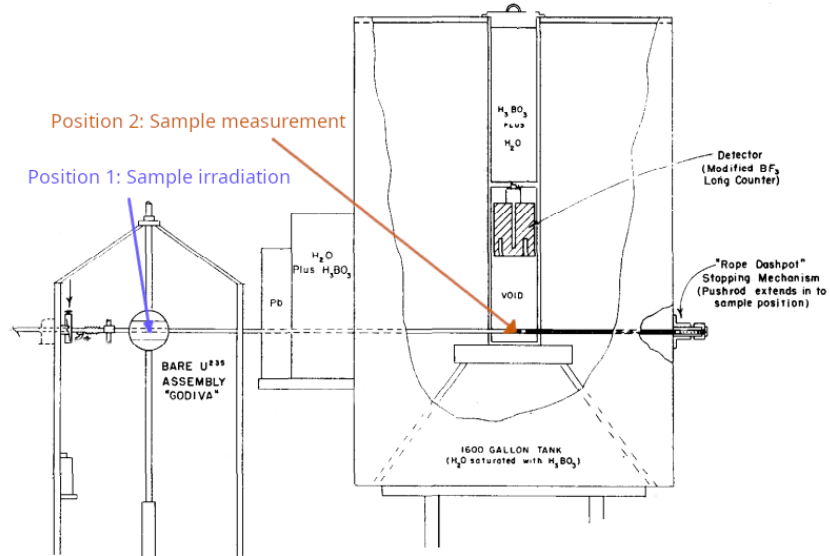


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [84].

Challenges associated collecting this experimental data [10] include experimental noise and signal obfuscation

due to moderation. Experimental noise becomes significant after long periods of time relative to the half-life of the longest-lived DNP, around 1 minute from ^{87}Br , as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation, so there is a gap between irradiation and when the sample can be moved to a delayed neutron counter. Even after a pulse irradiation, the long-lived groups form a background that interfere with measurement of the short-lived groups [10]. Due to these effects, measurement of delayed neutrons from the sample is less accurate initially, becomes increasingly accurate, and then becomes less accurate again as the delayed neutron counts decrease.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, the fission chamber does not account for source multiplication within the irradiated sample. Keepin et al. suggests a technique to correct for the sample self-multiplication. The sample self-multiplication require a correction factor of up to 3% in the number of fissions.

By the year 2000, there were already 245 individual sets of DNP group parameters for 20 different fissile nuclides [168]. Spriggs and Campbell go into a high level of detail, discussing the various analysis methods, neutron energies, fissile nuclides, counting intervals, and sample compositions for many of these DNP group parameter formulations [168]. Tuttle also provides a detailed review of many macroscopic DNP group parameters pre-1974, including details on the sample, reactor, counter, and travel time of the sample between irradiation and measurement [177]. Tables 2.1 through 2.4 show some macroscopic DNP group parameters from the literature for thermal fission of ^{235}U [123].

Table 2.1: Macroscopic formulation of group yields from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [84]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [34]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
Saleh et al. [155]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [171]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [177]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[184]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070

* Modified to include a sixth group by Parish et al. [123].

Table 2.2: Macroscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [84]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [34]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
Saleh et al. [155]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [171]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [177]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[184]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [123].

There is reasonable agreement between the different DNP group parameters, though there is variation among

Table 2.3: Macroscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [84]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [34]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
Saleh et al. [155]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [171]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [177]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[184]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911

* Modified to include a sixth group by Parish et al. [123].

Table 2.4: Macroscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [84]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [34]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
Saleh et al. [155]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [171]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [177]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[184]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545

* Modified to include a sixth group by Parish et al. [123].

them. Not only did each approach use a slightly different experimental methodology via different energy spectra, data collection methods, sample transfer time between irradiation and measurement, sample size, and sample composition, but there are also differences in how the data was processed. There are several ways that the data was processed from delayed neutron count rates into DNP group parameters among the works in these tables: simultaneous non-linear least-squares fitting of saturation and pulse irradiation data, separate non-linear least-squares fitting of pulse and saturation irradiation data and subsequent combination, non-linear least-squares of an intermediate amount of time between a pulse and saturation, and exponential stripping.

Keepin et al. used both a simultaneous fitting of DNP group parameters from pulse and saturation irradiation data as well as a separate fitting of four groups from each irradiation. The separate fitting of four groups were normalized at the second group, and Keepin et al. reports that this method gave better results than the simultaneous fit [84].

Charlton et al. use the equation for an intermediate amount of time that is not infinite or a pulse, as shown in Equation (2.5). They also only use five groups, as the sample took half a second to move from irradiation to the detector where delayed neutron counts could be measured. The shortest lived group six DNPs were not included, but Parish et al. added group six values for their own analysis [123]. Charlton et al. use a least-squares solve to determine the DNP group parameters, according to Spriggs and Campbell [168].

Saleh et al. use the equation for an intermediate amount of time, similar to Charlton et al., where the equation is described in Equation (2.5). However, instead of using a least-squares solve, they use exponential stripping [155].

Synetos and Williams use the same approach as Charlton et al. [171, 169].

Tuttle formulates suggested values based on reviewing the literature and applying empirical correlations [177].

Waldo et al. apply a transit time correction using an empirical correlation, and the DNP group parameters are fitted using a least-squares solve [184]. However, it is not made clear if pulse or saturation irradiations are used. Because the conclusions indicate that the DNP group parameters are less applicable for short times or short irradiations, this implies that a saturation irradiation was used.

Microscopic Measurement of DNPs

The other approach used to measure DNPs is the microscopic – also referred to in the literature as the aggregate or summation – approach, which uses data from individual nuclides summed together to form a simulated delayed neutron count rate. This count rate can then be used to generate DNP group parameters, the same as in the count rate from the macroscopic method. This approach was investigated in detail in the 1990-2000's [131, 41, 151, 187, 22], with progress through the 2010's with more data [114, 76]. Leconte et al. used this approach as recently as 2020 [95]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape to rapidly measure short lived nuclides, such as in the study by Björnstad et al. where an aluminized mylar tape is used [18]. The magnetic tape is used to transport samples or remove longer-lived decay products [163, 152]. Experimental methods include the neutron- β coincidence technique, separate counting of neutrons and β particles, separate counting of neutrons and γ rays, ion counting, fission yield from a set number of fissions, gamma counting of a long-lived daughter, and normalization of the neutron counter using a known precursor [152].

Other experimental methods used to study DNPs include on-line mass spectrometry, helium-3 spectrometry, proton-recoil detectors, the time-of-flight method, and neutron scintillators [42]. On-line mass spectrometry combined with a helium-3 ionization chamber and a proton-recoil detector allows for measurement of the delayed neutron energy spectrum [162, 17]. The time-of-flight technique consists of measuring the flight time of a neutron over a known distance to calculate the velocity, including a neutron scintillator [17, 38, 60, 37].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. The IAEA has compiled and evaluated this data in a publicly accessible form [47, 46]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly rather than using groups. This is particularly useful for understanding what nuclides have the greatest impact, as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

Tables 2.5 through 2.8 show some microscopic DNP group parameters from the literature for thermal fission of ^{235}U [123]. There is reasonable agreement between the different DNP group parameters, though it should be noted that Brady and England may be closer to ENDF/B-VI.8 because ENDF/B-VI.8 uses the Brady and England results in its formulation.

ENDF-B/VI.8 pulls from Brady and England, as well as other works by England [148, 22, 50, 23]. This work and Brady and England's approach entails collecting 271 DNPs, generating a delayed neutron count rate curve, and then using the two four group aggregation approach described by Keepin et al. [84, 22].

Mills et al. use two different approaches: fitting all parameters simultaneously as described by Keepin et al. and using the Keepin et al. half-lives while fitting only the yields [115, 84]. The values presented here are from the approach where all parameters are fit simultaneously for all DNP group parameters.

Table 2.5: Microscopic formulation of group yields from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [148]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [115]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [22]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158

* Modified to include a sixth group by Parish et al. [123].

Table 2.6: Microscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [148]	-	-	-	-	-	-
Mills et al. [115]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [22]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [123].

Table 2.7: Microscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [148]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [115]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [22]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462

* Modified to include a sixth group by Parish et al. [123].

Table 2.8: Microscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [123]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [148]	-	-	-	-	-	-
Mills et al. [115]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [22]	-	-	-	-	-	-

* Modified to include a sixth group by Parish et al. [123].

Comparison of Macroscopic and Microscopic Approaches

Huynh and Jouanne offer a very useful comparison between the macroscopic and microscopic approaches via different datasets [76]. Specifically, they compare the total delayed neutron yield from various data sources from macroscopic measurements to the calculated total delayed neutron yield from microscopic measurements. This data can also be compared to the previously shown tables, from which the total delayed neutron yield can be calculated. The total data compilation is presented in Table 2.9.

Table 2.9 shows that the total delayed neutron yield for thermal fission of ^{235}U ranges from 0.01477 to 0.01905

Approach	Data	Delayed Yield
Microscopic	JEF-2.2 [10]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [156]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [33]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [148]	1.670
Microscopic	Mills et al. [115]	1.645 ± 0.07
Microscopic	Brady and England [22]	1.780 ± 0.05
Macroscopic	JEF-2.2 [10]	1.655
Macroscopic	JEF-3.1.1 [156]	1.620
Macroscopic	ENDF/B-VII.0 [32]	1.585
Macroscopic	Keepin et al. [84]	1.580 ± 0.07
Macroscopic	Charlton et al. [34]	1.577 ± 0.03
Macroscopic	Saleh et al. [155]	1.645 ± 0.11
Macroscopic	Synetos and Williams [171]	1.510 ± 0.06
Macroscopic	Tuttle [177]	1.700 ± 0.05
Macroscopic	Waldo et al.[184]	1.670 ± 0.07

Table 2.9: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [76]).

(or 100 times that for 100 fissions, which makes the data more legible). The microscopic values range from 1.477 to 1.905, capturing the minimum and maximum of the range of values seen in the table. The macroscopic values range from 1.510 to 1.700, presenting a tighter range of values. Another point of interest is that there does not appear to be a consistent difference between the microscopic and macroscopic values for the same dataset. For instance, JEF-2.2 has a larger microscopic yield, but JEF-3.1.1 has a larger macroscopic yield.

2.1.2 DNP Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally understanding delayed neutron precursors were discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The microscopic and macroscopic experimental approaches provide the necessary data to create more easily handled forms. By summing the contributions from each DNP in the microscopic approach, the macroscopic and microscopic approaches can be said to provide data in the form of a delayed neutron count rate over time. Section 2.1.2 discusses the least-squares approach to convert this data into several DNP groups.

The DNPs are commonly represented in six groups. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and England found that increasing the number of groups for point kinetics calculations from six to nine caused no significant differences [22]. Their study shows that increasing the number of DNP groups alone does not provide an increase in accuracy for point kinetics simulations.

Representing the DNPs in group form also simplifies the calculation of various kinetics parameters. Various kinetics parameters are introduced in Section 2.1.3, though without detail on the purpose of each variable. This section provides more information on what these kinetics and delayed neutron group parameters physically represent, while Sections 2.1.2 and 2.1.2 detail how they are calculated. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.10 goes through each of the terms of importance for this section.

Parameter	Description
λ_k / λ_i	The decay constant for DNP group k / nuclide i
$\bar{\nu}_{d,k} / \bar{\nu}_{d,i}$	The delayed neutron yield for DNP group k / nuclide i
χ_k / χ_i	The spectra of DNP group k / nuclide i
$P_{n,i}$	The delayed neutron emission probability for nuclide i
$\langle T \rangle$	The mean half-life
Y	The total delayed neutron yield
β_k / β_i	The delayed neutron fraction of DNP group k / nuclide i
β	The total delayed neutron fraction
β_{eff}	The effective delayed neutron fraction
Λ	The mean neutron generation time
Λ_{eff}	The effective mean neutron generation time

Table 2.10: DNP group parameters and kinetics parameters along with their definitions.

The DNP groups come with associated parameters, such as the group decay constants, λ_k , and the group yields, $\bar{\nu}_{d,k}$. These group yields can be converted to group delayed neutron fractions, β_k , by dividing the group yields by the average number of neutrons emitted per fission event. The group yields represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_k and λ_k terms are the defining parameters for the DNP groups, along with the group spectra, χ_k , for DNP group k . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [129]. In 1989, Brady and England also use the individual nuclides to compare with group data [22], and updated the six-group parameters based on the explicit DNPs. Updating the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. The effective multiplication factor, k_{eff} , is an important term in reactor physics. This term represents the scaling of the number of neutrons over each generation in a finite reactor system. The infinite multiplication factor, k_{∞} , represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction and mean generation time terms, where the effective form of the term requires information on the specific reactor geometry.

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance, or weighting, function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection. The adjoint flux can be calculated by taking the adjoint of the Boltzmann transport equation, or by using the iterated fission probability (IFP) method, which provides an importance term proportional to the adjoint flux [174, 118].

The delayed neutron fraction, β , represents the ratio of delayed neutron production over all neutron production [113]. Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the delayed-neutron induced fission to the total number of fission events weighted by the spectrum and adjoint neutron flux [147]. Alternatively, it can be thought of as the ratio of the spectrum weighted and adjoint weighted delayed neutron production over all spectrum weighted and adjoint weighted neutron production [113]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronic-important regions.

Another term, the mean neutron generation time, Λ , is the mean time between generation of a neutron and fission induced by that neutron [147]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [179]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [179].

Calculation of DNP Group Parameters

The DNP group parameters are important for simulating transients. The DNP group parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.2) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for I individual DNPs for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t).$$

Equation (2.3) instead gives the K group delayed neutron count rate by the same detector for a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \tilde{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.3)$$

For an infinite irradiation (also referred to as a saturation irradiation), Equation (2.4) is used rather than Equation (2.3), given as [24]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \tilde{v}_{d,k} e^{-\lambda_k t}. \quad (2.4)$$

Equation (2.5) shows the intermediate case, where the irradiation time, T , is neither infinite nor a pulse [187]. This equation gives the delayed neutron count rate as:

$$\dot{n}_d(t, T) \approx \epsilon \dot{F}_s \sum_{k=1}^K \tilde{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}). \quad (2.5)$$

The terms in Equations (2.3), (2.4), and (2.5) are defined as follows:

- K = the number of DNP groups [-]
- λ_k = the decay constant of DNP group k [s^{-1}]
- F_s = the total number of fission events in the pulse irradiation [-]
- $\dot{F}_s$ = the fission rate for a saturation irradiation [s^{-1}]
- $\bar{v}_{d,k}$ = the group yield for group k [-] .

More information on the derivation of these equations can be found in Appendix A. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [187].

The pulse irradiation approach gives better statistics on the short lives DNPs, while the longer lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [24]. Specifically, Brady uses the four long-period groups from the saturation irradiation and the four short-period groups from the pulse irradiation renormalized at the second DNP group.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays. For example, Brady used STEPIT to handle the non-linear least-squares solve with both pulse and infinite irradiation experimental data [24]. Waldo et al. also solve the least-squares problem, though the specific approach used is not specified [184]. The general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.6)$$

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.3) and (2.4). Equation (2.7) gives the coefficient matrix for a pulse irradiation as:

$$A = \epsilon F_s \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{bmatrix}, \quad (2.7)$$

though for a saturation irradiation the only change would be removing the λ_i term.

Within this least squares equation, the solution vector contains only the group yields, given in Equation (2.8) as:

$$\vec{x} = \begin{bmatrix} \bar{v}_{d,1} \\ \bar{v}_{d,2} \\ \vdots \\ \bar{v}_{d,K} \end{bmatrix}. \quad (2.8)$$

The target vector contains the linearized delayed neutron count rate data normalized to the number of fissions, or fission rate, and the detector efficiency, as shown in Equation (2.9) as:

$$\vec{b} = \begin{bmatrix} \ln(n_d(t_0)) \\ \ln(n_d(t_1)) \\ \vdots \\ \ln(n_d(t_m)) \end{bmatrix}. \quad (2.9)$$

The natural logarithm is needed here, as otherwise the short time steps will dominate the group parameters. Another way to resolve this is to use a relative residual, as given by Equation (2.10).

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b} + \epsilon} \right\|, \quad (2.10)$$

where,

$$\begin{aligned} \epsilon &= \text{a small value to avoid division by zero} \\ &= 1 \times 10^{-12}. \end{aligned}$$

By solving this non-linear system of equations, the group half-lives, T_k , and yields, $\bar{v}_{d,k}$, can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, F , then Equation (2.11) can be used to determine the K group half lives using [42] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.11)$$

where,

$$\begin{aligned} \lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F. \end{aligned}$$

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Equation (2.12) shows that including the energy in the delayed neutron count rate requires the energy spectra, $\chi_{d,k}(E)$, of the relevant group to be included. This is the general form of the saturation and pulse irradiation equations, where A_k represents $\bar{v}_{d,k} \lambda_k$ for a pulse irradiation or $\bar{v}_{d,k}$ for saturation irradiation. $\tilde{F}_s$ is the total number of fissions for a pulse irradiation or the fission rate for saturation irradiation. The generic energy and time dependent neutron count rate is given as:

$$\dot{n}_d(E, t) \approx \epsilon \tilde{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.12)$$

where,

$$\begin{aligned} \tilde{F}_s &= \text{the generic fission term [- or } s^{-1}] \\ A_k &= \text{the generic matrix term [} s^{-1} \text{ or -]} \\ \chi_{d,k}(E) &= \text{the delayed neutron energy spectrum of DNP group } k \text{ [-]}. \end{aligned}$$

Calculation of DNP Group Spectra and Concentrations

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is important for situations where reducing the problem from several hundred DNPs to a few DNP groups reduces computational cost. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group

spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

Two different methods are considered for separating and combining the DNP groups into individual nuclides and vice versa. These methods have been used to generate DNP group spectra, so the group spectra can also be calculated using these methods:

- Grouping based on half lives [51, 149]
- Grouping based on fractional contribution to adjacent groups [24]

The grouping based on half lives requires comparing each nuclide's half life to a constant value. The groupings based on half lives are given in the top two sets in Table 2.11. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach [28].

Table 2.11: Half-lives of DNP groups from various sources.

Source	T_1 [s]	T_2 [s]	T_3 [s]	T_4 [s]	T_5 [s]	T_6 [s]
[51]	$T > 31$	$31 \geq T > 9$	$9 \geq T > 3$	$3 \geq T > 0.8$	$0.8 \geq T > 0.3$	$0.3 \geq T$
[149] [*]	$T > 30$	$30 \geq T > 10$	$10 \geq T > 4$	$4 \geq T > 1.4$	$1.4 \geq T > 0.4$	$0.4 \geq T$
[28]	$T = 51.98$	$T = 21.17$	$T = 5.74$	$T = 2.29$	$T = 0.82$	$T = 0.24$

^{*} Rudstam uses $T_1 = 55.6$, but to ensure all DNPs are included, this has been modified.

The half-life grouping method thus results in a series of comparisons; if a nuclide is within a given group's bounds, the concentration of that nuclide is added to that group.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups, i and $i + 1$, for each nuclide k . These groups are determined via Equation (2.13) as:

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.13)$$

Nuclides which don't fit in these bounds will have the entire fractional contribution go to the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.14) as:

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.14)$$

To solve for the fractions, the least squares minimization problem is solved as:

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.15)$$

Solving this equation for where the partial derivatives are zero for each fraction, $f_{k,i}$ and $f_{k,i+1}$, combined with

Equation (2.14), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i\lambda_{i+1}}{\lambda_i+\lambda_{i+1}} \\ \frac{2\lambda_i\lambda_{i+1}}{\lambda_i+\lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} \frac{2\lambda_i\lambda_k}{\lambda_i+\lambda_k} \\ \frac{2\lambda_k\lambda_{i+1}}{\lambda_k+\lambda_{i+1}} \\ 1 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix}. \quad (2.16)$$

Because the number of rows is greater than the number of columns in the matrix, the left-inverse is used. This system of equations is then solved as shown in Equation (2.17) as:

$$\vec{x} = (A^T A)^{-1} A^T \vec{b} \quad (2.17)$$

With the fractional contribution terms, $\vec{x}$, the concentration of a given nuclide can then be multiplied by each fraction to show how much it contributes to that group. With a summation of each nuclides contribution, the group concentrations can thus be calculated and vice versa. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

Calculation of Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The k^{th} DNP group delayed neutron fraction is calculated as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}. \quad (2.18)$$

A useful term for comparing DNP group parameters is the average half-life [132, 133]. Parish et al. also use the average half-life, but do not include the natural log of two in their equation, which could be a typo [123]. Equation (2.19) gives the average half-life as:

$$\langle T \rangle = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_k}. \quad (2.19)$$

Equation (2.20) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event, $\bar{v}_d$, can be calculated [64]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.20)$$

This yield can also be approximately represented for an individual nuclide by integrating Equation (2.1) over all time, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0)e^{-\lambda_i t}. \quad (2.21)$$

This yields Equation (2.22), where the total number of delayed neutrons emitted from a single nuclide i is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.22)$$

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{v}_d}{\bar{v}} \quad (2.23)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.24)$$

where,

$$\begin{aligned} l &= \text{the prompt neutron lifetime [s]} \\ k &= \text{the neutron multiplication factor [-]}. \end{aligned}$$

(2.25)

This can be calculated using a time-dependent version of a Monte Carlo code along with point reactor kinetics (PRK) [147].

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} is the spectrum and adjoint weighted ratio of the delayed neutron induced fission to the induced fissions from all neutrons [147]. This is given in Equation (2.26) as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}, \quad (2.26)$$

where,

$$\begin{aligned} \Phi^\dagger &= \text{the adjoint neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\ \chi_d &= \text{the delayed neutron energy spectrum [-]} \\ \chi &= \text{the neutron energy spectrum [-]} \\ \beta &= \text{the delayed neutron fraction [-]} \\ \bar{v} &= \text{the average number of neutrons released per fission [-]} \\ \Sigma_f &= \text{the macroscopic fission cross section } [cm^{-1}] \\ \Phi &= \text{the neutron flux } [n \cdot cm^{-2} \cdot s^{-1}]. \end{aligned}$$

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.27) as [179]:

$$\Lambda_{eff} = \frac{\int \Phi_0^\dagger \frac{1}{\bar{v}} \Phi_0 dE d\vec{\Omega} d\vec{r}}{\int \Phi_0^\dagger F_0 \Phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.27)$$

where,

$\Phi_0^\dagger$ = the critical adjoint neutron flux [$n \cdot cm^{-2} \cdot s^{-1}$]

Φ_0 = the critical neutron flux [$n \cdot cm^{-2} \cdot s^{-1}$]

$\bar{v}$ = the average neutron speed [$cm \cdot s^{-1}$]

F_0 = the total fission operator [cm^{-1}].

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [179]. These include the direct integration, or adjoint weighted, method [85]; the eigenvalue, or perturbation, method [69]; and the iterated fission probability method [19]. Λ_{eff} can also be calculated using the pulsed method [147]. For β_{eff} , there are also the prompt, Meulekamp, and Spriggs methods which can be used [147, 113]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [8].

2.1.3 Kinetics and Dynamics Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [188]. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena when generating the DNP group parameters.

Multi-Group Diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches applied to transient problems, surpassed only by transport methods. Hurwitz and Ehrlich provide a standard formulation of multi-group diffusion written in 1953 [75].

Examples of software that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [56, 172]. Other examples can be found in the review by Wooten et al. [88, 57, 87] and elsewhere in the literature [192, 190, 89, 119, 125]. These studies typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling is given as [188]:

$$\begin{aligned} \frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_{g'}, \end{aligned} \quad (2.28)$$

and,

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\bar{v}_g \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i, \quad (2.29)$$

where,

- $\bar{v}_g$ = the average number of neutrons produced by fission of group g [-]
- ϕ_g = the scalar neutron flux of group g [$n \cdot cm^{-2} \cdot s^{-1}$]
- D_g = the group g diffusion term [cm]
- G = the total number of energy groups [-]
- $\Sigma_{s,g' \rightarrow g}$ = the macroscopic scattering cross section from group g' to group g [cm^{-1}]
- $\chi_{p,g}$ = the number of prompt neutrons born in group g [-]
- $\chi_{d,g,i}$ = the number of delayed neutrons from precursor group i born in group g [-]
- C_i = the concentration of DNPs in group i [$atoms \cdot cm^{-3}$]
- $\vec{u}$ = the velocity profile [$cm \cdot s^{-1}$]
- $\Sigma_{a,g}$ = the macroscopic absorption cross section for group g [cm^{-1}]
- ν_T = the eddy viscosity [$cm^2 \cdot s^{-1}$]
- Sc_T = the turbulent Schmidt number [-].

Modified Point Kinetics

PRK are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t), \quad (2.30)$$

and,

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t), \quad (2.31)$$

where,

- n = the relative neutron density [-]
 - ρ = the reactivity [-]
 - β_{eff} = the effective delayed neutron fraction [-].
- (2.32)

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [69, 92]. Modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [188]. Wooten et al. provide four different publications which use MPK in MSR analysis [65, 112, 165, 191].

The work by Guo et al. analyzes unprotected loss of heat sink (ULOHS) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified

into the delayed point reactor kinetics (DPK), which model precursor concentrations in-core and ex-core, with the primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,c}(t), \quad (2.33)$$

$$\begin{aligned} \frac{dC_{i,c}(t)}{dt} = & \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_c} \right) C_{i,c}(t) \\ & + \frac{1}{\tau_l} \left(\frac{V_l}{V_c} \right) C_{i,l}(t - \tau_l), \end{aligned} \quad (2.34)$$

$$\frac{dC_{i,l}(t)}{dt} = \frac{1}{\tau_c} \left(\frac{V_c}{V_l} \right) C_{i,c}(t) - \left(\lambda_i + \frac{1}{\tau_l} \right) C_{i,l}(t). \quad (2.35)$$

These equations, shown in Equations (2.33) through (2.35) (recreated from [65, 165]), are also used by Shi et al. in the Reactor Excursion and Leak Analysis Program (RELAP). Shi et al. simulate a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [165]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but fails to capture all oscillatory effects [188].

Intermediate terms used in Equations (2.33) through (2.35) are shown in Equations (2.36) through (2.39) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \quad (2.36)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,l}, \quad (2.37)$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \quad (2.38)$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}. \quad (2.39)$$

The terms in Equations (2.33) through (2.39) are defined as follows:

$$\begin{aligned} V_l &= \text{the volume of the loop [cm}^3\text{]} \\ V_c &= \text{the volume of the core [cm}^3\text{]} \\ \beta_{static} &= \text{the value of } \beta_{eff} \text{ without flow [-]} \\ M_c &= \text{the mass in the core [kg]} \\ M_l &= \text{the mass in the loop [kg]} \\ \dot{M}(t) &= \text{the mass flow rate [kg} \cdot \text{s}^{-1}\text{]}. \end{aligned}$$

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [170]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses the "IPK method," which may stand for "Improved Point-Kinetic method," though this is not explicitly stated [112]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [112].

The work by Zanetti et al. extends the TRACE code, a finite volume code used for LWR transients [191]. The

DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [93]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich and used in PRK equations [68].

Quasi-Statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [188]. The general formulation used for quasi-static methods is very similar among the various publications that use it [48, 172]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [69, 45]. There are also other studies which discuss the quasi-static method in detail [12].

Wooten et al. introduce several different publications which have used quasi-statics [188]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [141]. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used. The neutron flux is treated as:

$$\Phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.40)$$

where,

$$\Phi = \text{the neutron flux } [n \cdot cm^{-2} \cdot s^{-1}]$$

$$\phi_A = \text{the neutron flux amplitude function } [n]$$

$$\psi(\vec{x}, t) = \text{the neutron flux shape function } [cm^{-2} \cdot s^{-1}].$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.41) and (2.42). These equations are solved by time-discretizing Equation (2.41) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5]. These primary equations are given as:

$$\begin{aligned} \frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v}\phi_A} \frac{d\phi_A}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta)\chi_p F(t)\psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t), \end{aligned} \quad (2.41)$$

and,

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = \phi_A \beta_i F \psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t) C_i(\vec{x}, t)). \quad (2.42)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters and flux amplitude are calculated. The equations used in the "reactivity" steps are recreated

here as:

$$\frac{d\phi_A}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}), \quad (2.43)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_i c_{s,i}. \quad (2.44)$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.45)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.46)$$

Intermediate terms used in Equations (2.43) through (2.46) are given as:

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla (u(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla (u(\vec{x}, t) C_{m,i}(\vec{x}, t)), \quad (2.47)$$

and,

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i C_{s,i}(\vec{x}, t). \quad (2.48)$$

The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.41) and (2.42) to zero as well as the fluid velocity to zero at $t=0$. This allows Equation (2.47) to be solved.

The terms in Equations (2.40) through (2.48) are defined as follows:

- $M(t)$ = the neutron disappearance operator [cm^{-1}]
- $F(t)$ = the neutron production operator [cm^{-1}]
- χ_p = the prompt neutron energy spectra [-]
- $C_i(\vec{x}, t)$ = the concentration of DNP group i [$atoms \cdot cm^{-3}$]
- $Q(\vec{x}, t)$ = the external neutron source [$n \cdot cm^{-3} \cdot s^{-1}$]
- $u(\vec{x}, t)$ = the DNP velocity [$cm \cdot s^{-1}$]
- $C_{s,i}(\vec{x}, t)$ = the "standard" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- $C_{m,i}(\vec{x}, t)$ = the "movable" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- ρ = the reactivity [-]
- $c_{s,i}$ = the PRK "standard" precursor concentration of group i [$atoms$]
- $c_{m,i}$ = the PRK "movable" precursor concentration of group i [$atoms$]
- q_{eff} = the PRK external neutron source [$n \cdot s^{-1}$]
- Λ = the neutron generation time [s]
- $W(x)$ = the weighting function (the default in SIMMER
is an adjoint flux computed at steady-state conditions) [-].

2.2 Molten Salt Reactor Fuel Depletion

There are many different MSR designs in the literature, as shown in Table 2.12 [36]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, causes depletion to vary spatially, leads to movement of DNPs, and requires close coupling of thermal-hydraulics to

neutronics for realistic modeling.

MSR Name	Source
FUJI series	[59]
Actinides Molten Salt TransmutER (AMSTER)	[180]
MOSART	[77]
Fast Spectrum Molten Salt Reactor (FS-MSR)	[71]
European MSFR	[111]
Small Mobile Molten Salt Reactor (SM-MSR)	[31]
MSBR	[143]
Molten Salt Reactor Experiment (MSRE)	[144]
Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor	[21]
Single fluid and two fluid denatured molten salt reactors (DMSRs)	[49, 4]
Thorium molten salt reactor (TMSR)	[121, 110]

Table 2.12: A collection of various MSR designs publicly available in the literature.

There are many publications in the literature which have investigated online chemical reprocessing of MSRs, as shown in Table 2.13 recreated from [158]. Table 2.13 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, showing most models do not include DNP movement in depletion. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models and is also sorted by the publication year in the "Year" column. The table shows there are many MSR concepts, shown in the "Reactor" column, for which depletion was modeled. The study by Valdez et al. investigates two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.13: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [158]).

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [80]	Continuous	Unit Cell	Several	No	2020
Zhuang et al. [194]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [189]	Continuous	Full 3D	MSFR	No	2019
Zhou et al. [193]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [9]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [145]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [154]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [16]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [140]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [124]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [164]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [120]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [55]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [117]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [110]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [66]	Mixed	2D Slice	MSBR	Yes	1971

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

2.2.1 Depletion Simulations

Depletion is the process of fission, transmutation, decay, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [97]. Simulating depletion entails numerically solving the Bateman equations; a generalized form of which is given as [97]:

$$\begin{aligned} \frac{dN_j}{dt} = & \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ & - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.49)$$

where,

- N_j = the atomic density of isotope j [$atoms \cdot cm^{-3}$]
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i [cm^2]
- ϕ = the spectrum-averaged scalar flux in the fuel region [$n \cdot cm^{-2} \cdot s^{-1}$]
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ [s^{-1}]
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ [cm^2]
- N_i = the atomic density of isotope i [$atoms \cdot cm^{-3}$]
- λ_j = the decay constant of isotope j [s^{-1}]
- $\lambda_{r,j}$ = the removal constant for removal of isotope j [s^{-1}]
- σ_j = the microscopic total transmutation cross-section of isotope j [cm^2]
- $\dot{N}_{f,j}$ = the feed rate for isotope j [$atoms \cdot cm^{-3} \cdot s^{-1}$].

The matrix form is given as:

$$\frac{d\vec{N}}{dt} = A\vec{N}, \quad (2.50)$$

where,

- $\vec{N}$ = the vector of material compositions [$atoms \cdot cm^{-3}$]
- A = the depletion matrix [s^{-1}].

The depletion matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well. Further documentation on these terms can be found in the OpenMC documentation, an open source Monte Carlo transport code which has built-in depletion solvers [146].

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [98]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [120] to implement 4th order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [9].

Overall, there are many approaches to solving the depletion equation with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.49) is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t), \quad (2.51)$$

where,

$$\begin{aligned}
S_j &= \text{the source term for nuclide } j, \\
&\quad \text{containing fission, decay, and transmutation sources } [atoms \cdot cm^{-3} \cdot s^{-1}] \\
\lambda_{net,j} &= \text{the loss term for nuclide } j, \\
&\quad \text{containing decay and transmutation losses } [s^{-1}].
\end{aligned}$$

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. The loss term, $\lambda_{net,j}$, does not vary with time in this equation.

This form of the Bateman equation does not have space or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel is more complex, as instead of solving a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.52)$$

where,

$$D = \text{the mass diffusivity } [cm^2 \cdot s^{-1}].$$

The depletion matrix is assumed to have no spatial resolution in this equation. Because of this assumption, only the concentrations and material blocks provide spatial resolution to the problem. Spatial refinement can be added to this equation based on the specific approach, such as using a spatially resolved neutron flux.

As previously mentioned, the main challenge faced when solving this problem is the computational cost. However, there are many studies which have attempted to develop methods to get around this problem. One common approach, the scaled flux method, is computationally inexpensive [15]. However, this method does not model the in-core and ex-core regions in a physical manner. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, and instead assumes the salt is perfectly mixed in the primary loop.

Another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [160].

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of the deterministic neutron transport code Griffin with the thermal fluids code Pronghorn, allowing for spatial and temporal resolution of certain nuclides [173]. Tano et al. present an approach where only insoluble, medium-lived, and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al., are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. has shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% compared to a spatially unresolved approach without flux scaling [173]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [173, 15]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

Numerical stability of the problem, when using an explicit-time solving scheme, uses the Courant number:

$$C = \frac{u\Delta t}{\Delta z}, \quad (2.53)$$

where,

Δt =the time step size [s]

Δz =the length between spatial nodes [cm].

The Courant number, C , must be less than or equal to 1 for the Courant–Friedrichs–Lewy condition in an explicit finite differencing scheme [159]. This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

For a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the computational cost is orders of magnitude smaller. A high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment is feasible to generate improved DNP group parameters.

2.3 Safety, Security, and Safeguards

2.3.1 Safety

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there are not specific events listed for MSRs, guidance is provided on categories of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, dependent events could potentially occur

at the same time.

Safety-related transients for MSRs include ULOHS, ULOF, unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [100, 40]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, causing a loss of cooling of the reactor. For the MSFR, Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [26]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results.

2.3.2 Security

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [137]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [6, 61]. There are several different categories which are important to understanding security: the actor type, the actor capabilities, their objectives, and the strategies they use to achieve said objectives. The main consideration for security in this work is protection of the public by preventing: theft of hazardous radioactive material and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [137].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

There are five categories of actor capabilities: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect their strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant more tools accessed with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including

cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, the specific actor capabilities and strategies are not considered. Additionally, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. This work investigates the formulation of new DNP group parameters, which impact the reactor behavior during a transient. There may be security issues that arise during a sabotage-initiated transient due to the recalculated kinetics and DNP group parameters.

2.3.3 Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [137]. This definition agrees with the definition made at the IAEA international workshop in 2002 [78]. In this work, proliferation resistance (PR) is used as a way to understand and measure certain criteria used in safeguards. PR is a spectrum, so it should not be considered necessary to optimize the PR, but rather as a way to evaluate strengths and weaknesses in the design relating to safeguards.

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [137]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [137].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of 2023 that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [72]. Based on the more recent Order, special nuclear material (SNM) is classified as shown in Table 2.14.

Table 2.14: The SNM classification table is recreated from [72].

Material Type	Accountable Quantity	Weight Field Used for Element	Weight Field Used for Isotope
^{235}U	1 g	total uranium	^{235}U
^{233}U	1 g	total uranium	^{233}U
^{242}Pu	1 g	total plutonium	^{242}Pu
$^{239-241}\text{Pu}$	1 g	total plutonium	$^{239}\text{Pu} + ^{241}\text{Pu}$
^{238}Pu	0.1 g	total plutonium	^{238}Pu
$^{241}\text{Am}^{**}$	1 g	total americium	^{241}Am
$^{243}\text{Am}^{**}$	1 g	total americium	^{243}Am
$^{237}\text{Np}^{**}$	1 g	total neptunium	-

^{**} Americium and neptunium are treated as SNM if they are separated.

Table 2.14 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". For this work, only materials of attractiveness level "C" and below are relevant. High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% ^{235}U or <10% ^{233}U .

The quantities provided in Table 2.14 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or $^{233}\text{U} \geq 6$ kg is considered Category I SNM. Alternatively, if the same material contains ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am , then an amount ≥ 20 kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" do not have a listed Category I quantity for material containing ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am .

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). SQs of the different material categories are given in Table 2.15 [137, 3, 11].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine

Use	Material	Significant Quantity [kg]
Direct	Plutonium	8
Direct	HEU	25
Direct	²³³ U	8
Indirect	²³⁵ U	75
Indirect	Natural Uranium	10,000
Indirect	Depleted Uranium	20,000

Table 2.15: Significant quantities of various materials are given along with their use category.

the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

Chapter 3

Methodology

3.1 Tools

Several external software tools evaluate and generate the improved DNP group parameters. Molten Salt Delayed Neutron (MoSDeN), an in-house developed Python package, collects and combines data, OpenMC is used for data handling and higher fidelity analysis, and the MOOSE app Moltres evaluates and refines them at different fidelity levels [178, 146, 102]. Each tool is described in detail in its corresponding sub-section.

3.1.1 MoSDeN

MoSDeN is a Python package designed to enable generation of DNP group parameters using the microscopic approach for any given reactor of interest. MoSDeN allows for varying levels of fidelity in the generation, ranging from only using nuclear data and broad assumptions to a high-fidelity multiphysics simulation with neutronics and thermal-hydraulics. The MoSDeN package relies on the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [182, 67, 74, 175, 53]. Many features of MoSDeN also require OpenMC to function properly [146]. MoSDeN automates data collection and processing, simulates pulse and/or saturation irradiations, converts the concentrations into a delayed neutron count rate, and generates group parameters.

There are three different levels of fidelity in MoSDeN: the 0D scaled model, the 0D flow model, and the spatially resolved flow model. Section 3.2 provides further details on the specific models incorporated or planned for MoSDeN.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutron and photon transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with MoSDeN, an in-house developed open-source Python package. The sample irradiation process is straightforward and configurable, as users can adjust parameters through an input file. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only depletion is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, MoSDeN by default uses the Evaluated Nuclear Data File (ENDF)/B-VII.1 based depletion chains provided by OpenMC and ENDF/B-VII.1 cross section data. The thermal spectrum chain applies to neutron

irradiation below 1 keV , while the fast spectrum chain applies to higher-energy neutron irradiation. The models by default assume monoenergetic neutrons irradiate the sample.

3.1.3 Moltres

Overview

MOOSE is an open-source, parallel, finite element framework for multiphysics applications [63]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [128, 101, 130, 126].

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and thermal-hydraulics equations [102]. This work uses Moltres to simulate the MSRE with different DNP group parameters. Moltres also validates the improved DNP group parameters in this work.

The MSRE experimental data can be used for validation, and was previously modeled with Moltres [102]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [126]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [125].

Moltres can simulate transients using the MSR-specific DNP group parameters, and it can generate improved parameters with higher fidelity than the other methods. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics.

Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.28), recreated here:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

Precursor Modeling

The time dependent DNP concentration is given in Equation (2.29) and recreated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{Sc_T} \nabla C_i$$

An issue arises with this equation because Moltres generates MSR-specific DNP group parameters, yet the equation requires predefined DNP group parameters. To generate the DNP group parameters, this equation is replaced by (2.52), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

Specifically, the solution to this problem is to use spatially resolved depletion over the irradiation simulation; instead of tracking concentrations of DNP groups, the simulation tracks concentrations of the individual DNPs.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow using Equations (3.1) and (3.2) recreated from [125]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{ref}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where,

ρ = fluid density

p = pressure

μ = dynamic viscosity

α = coefficient of thermal expansion

$\vec{g}$ = gravitational force vector

T = fluid temperature

T_{ref} = reference temperature at which the nominal density is provided.

The temperature distribution is given by Equation (3.3) recreated from [125], which shows the temperature advection-diffusion as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

and by Equation (3.4) recreated from [125], which gives the fission heat source term as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where,

c_p = specific heat capacity of molten salt

k = effective thermal conductivity of molten salt

Q_f = fission heat source

ϵ_g = average fission energy released by neutrons in group g ,

Q_s = heat sink/removal.

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomena to complete the model. This phenomena is the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to Equation (3.5), which describes the precursor concentration in the ex-core region as:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Generating DNP group parameter data in a reactor-agnostic manner requires a reactor-agnostic generation process. Fully detailing the flow regime with a large number of parameters is computationally infeasible. Instead, the flow is simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

The other required parameters describe chemical removal, which varies by fission product and reprocessing rate. Refueling is not considered in this work, as its effects are negligible over the short time frames analyzed. An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.1.6. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for analyzing the most extreme cycle times in this work. This is because the MSBR presents very large chemical removal rates, providing useful conservative estimates as to the impact these large removal rates could have on DNP group parameters. These chemical removal rates are given in Table 3.1 as cycle times, the inverse of the chemical removal rate. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have small removal rates relative to the time scales of interest for saturation irradiations. Section 4.1.6 examines these longer cycle times to confirm their negligible impact. Section 4.3 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Cycle Times from [143].

Reprocessing Group	Element(s)	Cycle Time (Full Power)
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	50 days
Rare Earths	Eu	500 days
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	20 seconds
Seminoble Metals	Zr, Cd, In, Sn	200 days
Gases	Kr, Xe	20 seconds
Volatile Fluorides	Br, I	60 days
Discard	Rb, Sr, Cs, Ba	3435 days
Salt Discard	Th, Li, Be, F	3435 days
Protactinium	Pa	3 days
Higher Nuclides	Np, Pu	16 years

With the chemical reprocessing scheme in place, the next step to generate the DNP group parameters is to decide on the level of fidelity. In this work, the three levels of fidelity are scaling terms to approximate different spatial volumes, accounting for flow using time differences and simulating the salt differently at different times where reprocessing can occur in different volumetric regions, and finally a model with spatial resolution where chemical processing can occur at specific spatial nodes.

3.2.1 0D Scaled Model

This is the lowest fidelity of the three models, and it primarily serves as a way to verify the microscopic approach is working as expected by comparing with existing literature. This approach can only model saturation concentrations, as there is no decay chain tracking nuclides decaying into each other. It is called the 0D scaled model because it uses the same concept from the scaled flux method, discussed in Section 2.2.2. Instead, the cumulative fission yield is used to calculate the equilibrium DNP concentrations, with a scaling term based on the amount of fuel salt in the core. An example solution with no reprocessing is shown in Equation (3.6) as [76]:

$$N_i = \frac{f_{in}CFY_i}{\lambda_i}, \quad (3.6)$$

where,

$$f_{in} = \text{the fraction of fuel-salt in the core}$$

$$CFY_i = \text{the cumulative fission yield of nuclide } i.$$

The derivation for this equation is given in Appendix B.

Another consideration is that no parasitic absorption is accounted for in this model, nor is time dependent decay of nuclides into DNPs. There is a possible approach to include the effects of parasitic absorption, but this

method is not included in this work [160, Benedict et al.].

Figure 3.1 shows the different data loaded into MoSDeN, including external data and user defined input data. The terms in green represent input data that should be considered constant for the DNP group parameters. The yellow terms indicate inputs that can be freely adjusted to formulate physically distinct DNP group parameters.

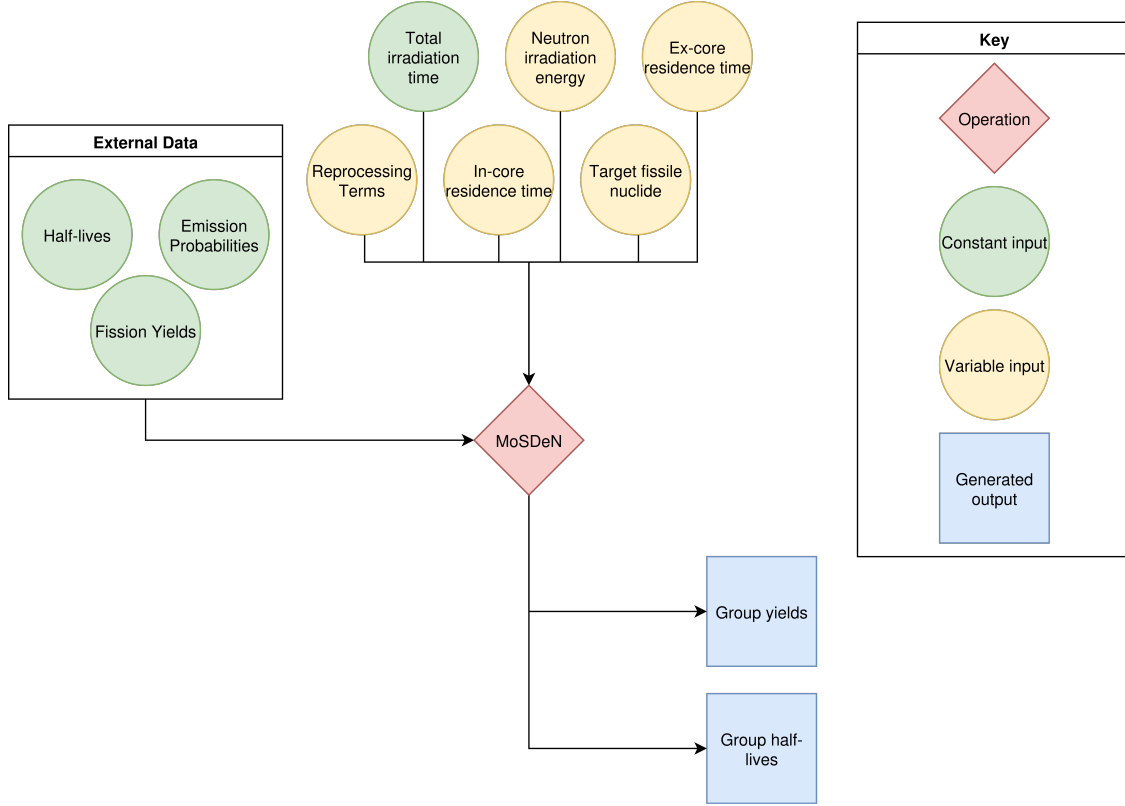


Figure 3.1: Flow diagram of 0D scaled model used to generate improved DNP group parameters.

After all this data is loaded into MoSDeN, it generates the concentration of each DNP. This concentration is then combined with emission probabilities and decay constants to form a delayed neutron count rate. A non-linear least squares solve then fits the improved DNP group parameters to the delayed neutron count rate, generating the group yields and half-lives, given as blue squares in Figure 3.1.

3.2.2 0D Flow Model

A simple model is necessary to generate sufficient data for each combination of input parameters, but there is still need for an increased level of fidelity compared to the 0D scaled model. The 0D flow model enables improved analysis of the general trends in nuclide evolutions. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times, and uses OpenMC to track the full depletion matrix. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least-squares problem, as shown in Appendix A.

Figure 3.2 shows how OpenMC uses this fission history to simulate irradiation on a control volume of fuel salt circulating in the primary loop. The model passes eight parameters into OpenMC [146]: in-core residence

time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. Three parameters, shown in green, remain constant: temperature, target density, and total irradiation time. The remaining inputs, shown in yellow, can be varied to generate physically unique DNP group parameters. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in MoSDeN. MoSDeN combines the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [47]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm [183, 25]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [84].

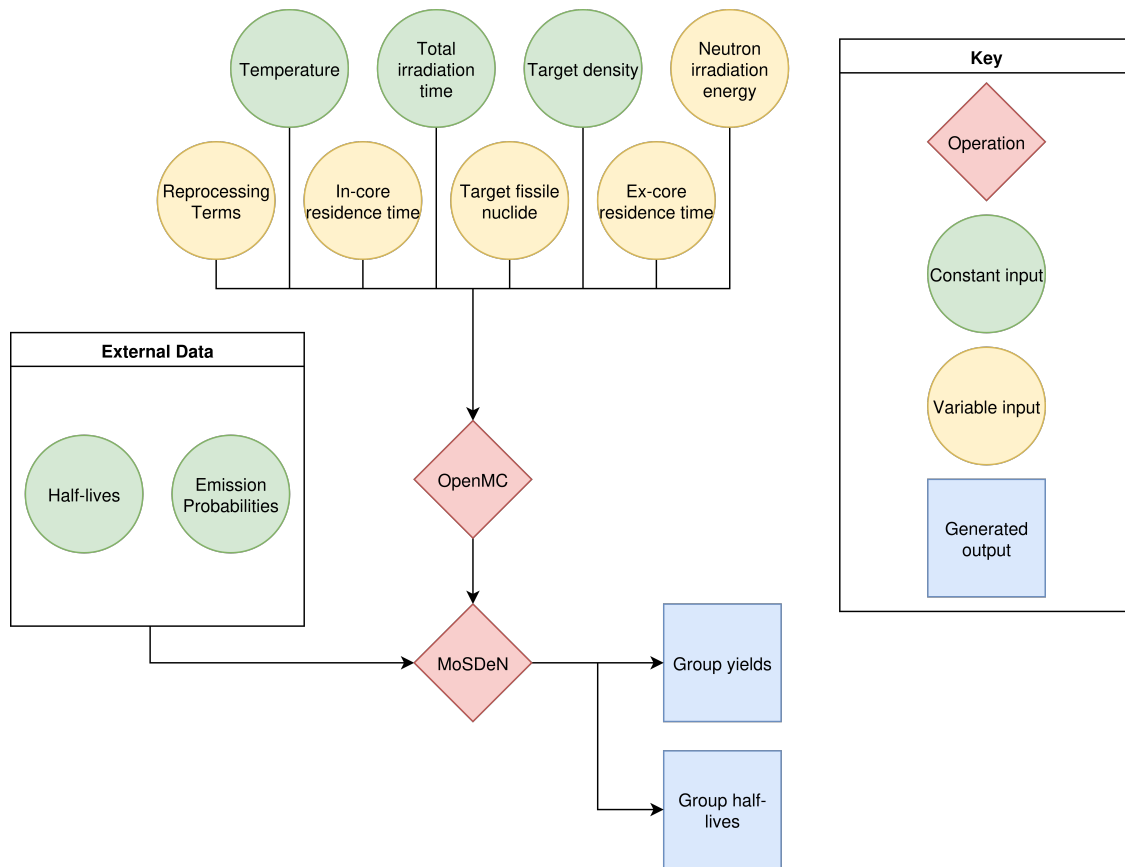


Figure 3.2: Flow diagram of 0D flow model used to generate improved DNP precursor group parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The exact neutron energy spectrum is not needed, as only significant differences in the neutron energy spectrum affect the DNP group parameters. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the OpenMC model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.7) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.7)$$

where,

- $\dot{S}$ = the source particle rate [$n \cdot s^{-1}$]
- S_{net} = the total number of source particles [n]
- τ_{in} = the in-core residence time [s]
- τ_{ex} = the ex-core residence time [s]
- T = the net irradiation time [s].

3.2.3 Spatially Resolved Model

The 0D flow model rapidly generates DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. Figure 3.3 shows the flow diagram for a spatially resolved model using Moltres instead of OpenMC.

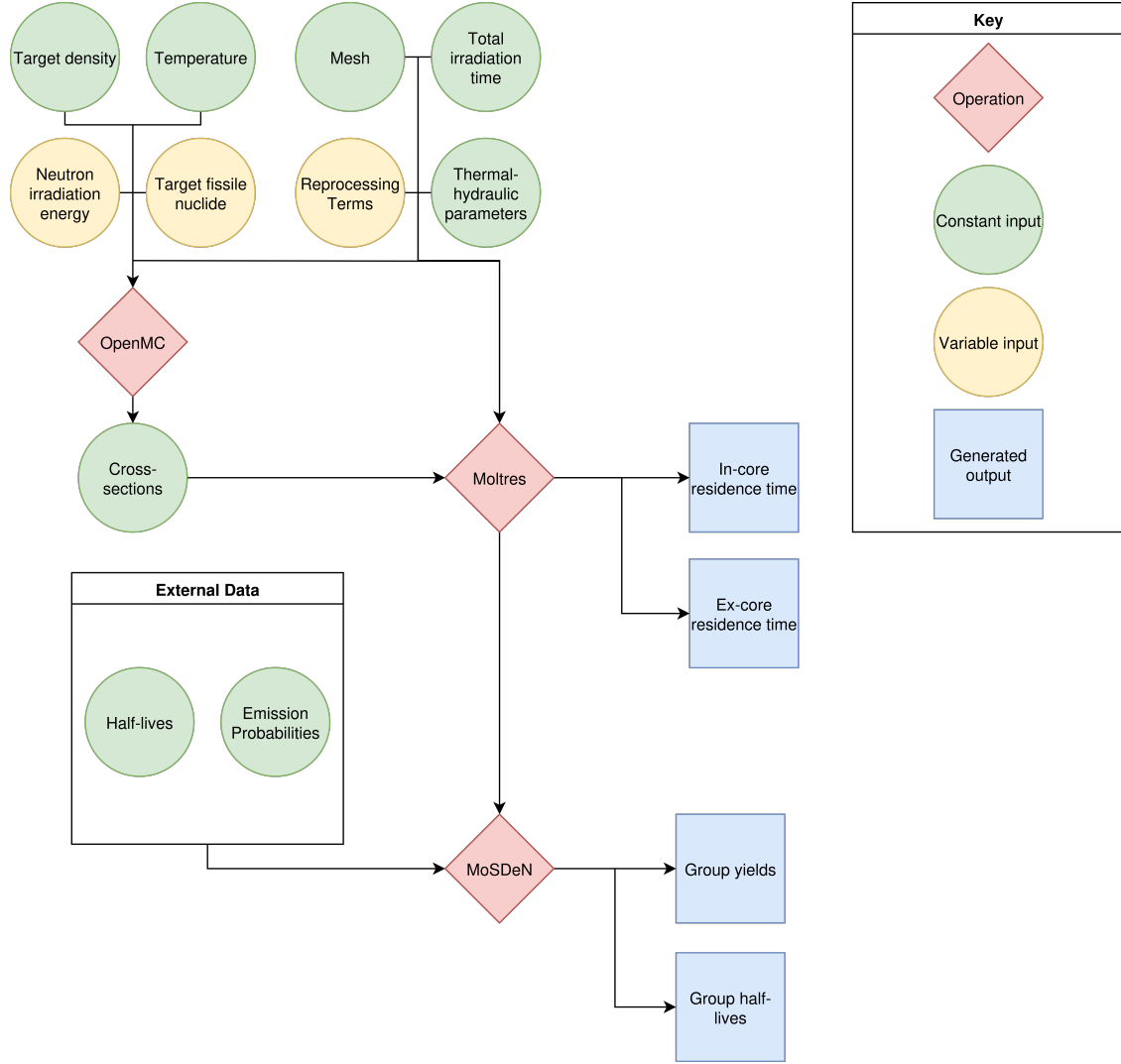


Figure 3.3: Flow diagram of spatially resolved flow model used to generate improved DNP group parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the problem geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [62]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.8) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, t) dt d\vec{x}}{\int_V \int_0^t N(\vec{x}, t) dt d\vec{x}} \tau_{net}, \quad (3.8)$$

where

$$\begin{aligned}
 V_{in} &= \text{the in-core volume [cm}^3\text{]} \\
 V &= \text{the total primary volume [cm}^3\text{]} \\
 N(\vec{x}, t) &= \text{the time and space resolved sample concentration [atoms} \cdot \text{cm}^{-3}\text{]} \\
 \tau_{net} &= \text{the net residence, or circulation, time of the fuel salt [s].}
 \end{aligned}$$

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.9) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.9)$$

3.3 Delayed Neutron Count Rate Generation

Although the 0D flow model overview has been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the OpenMC block. This includes a detailed discussion on the concentrations generated by OpenMC and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The OpenMC model uses a fixed-source problem on a three-gram spherical sample of a fissile target. A uniformly distributed source is placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy.

The fixed-source in this model consists entirely of neutrons at a specific energy, rather than a spectrum. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. However, this work assumes the spectral effects are sufficiently small to exclude. Photons can also generate delayed neutrons through photofission, though this is not investigated here [86].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \frac{\text{g}}{\text{cc}}$, the spherical sample has a radius of approximately 0.4 cm . The spatial resolution of concentrations within the sample is insignificant. Additionally, the concentrations extracted from the OpenMC simulation are spatially collapsed in the post-processing, so the spatial effect of a uniformly distributed source serves only to reduce any self-shielding effects which could reduce the statistics and cause more transmutation than expected.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are then converted into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, MoSDeN determines the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.10), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.10)$$

This is the approach used in the 0D scaled model.

This approach does not account for decay chains between nuclides. An alternative approach uses repeated zero-power depletion simulations in OpenMC to generate concentrations as a function of time, incorporating decay chains. This is used in the 0D flow model, though the 0D flow model also offers the option of using Equation (3.10). These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing MoSDeN to access more accurate concentrations at any time after irradiation [181]. This approach only models delayed neutrons generated during decay, as it is beyond the scope of this work to model other particles [35]. Section 4.1.7 compares these two approaches.

The concentrations are combined with Equation (2.2), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is converted into DNP group parameters. Fitting the DNP group parameters to the delayed neutron count rate curve requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.3), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the DNP group parameters. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [123]. Section 4.1.5 shows the sensitivity study for the time nodes, for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation than for the saturation irradiation. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produces worse fits, as the multiplication is not independent and is necessary for accurate modeling.

The residual of the solve is given in Equation (2.10), recreated here as:

$$r = \min_x \left\| \frac{b - Ax}{b + \epsilon} \right\|.$$

Half-life bounds are set from 1 millisecond to 1000 seconds, and the yield bounds are from 0 to 1. Convergence

tolerances for the cost function, change in independent variables, and gradient norm are all set to 1×10^{-12} to ensure tight convergence.

Sample self-multiplication, mentioned in Section 2.1.1, is another consideration. However, since the fission counts are taken directly from OpenMC tallies, self-multiplication is included and does not introduce error.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [53]. These uncertainties include uncertainties in fission yields, half-lives, and emission probabilities. The uncertainties are propagated through the count rate calculation, but for the non-linear least squares solve a Monte Carlo sampling based approach is used. Each count rate is generated using randomly sampled fission yields, half-lives, and emission probabilities within their uncertainty bounds for each DNP. This enables formation of a normal distribution of group parameters, from which the mean and standard deviation of the mean can be extracted.

Once the DNP group parameters and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [24]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients is evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.3. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated improved DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [84]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate

is accurately predicted, the formulation is considered valid. The second stage validates the improved DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [136]. This experiment has been studied using Moltres and Quasimolto, allowing comparison with their results [139, 125]. However, previous models excluded the inlet and outlet plena, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while improved parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 2.3.3, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [195]. The improved DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity responses than the improved parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied with a single significant quantity of material removed over the course of one year, simulated using OpenMC. The DNP group parameters are then generated for the fuel composition at that time, weighted by the volumetric fission rate fraction of each fissile nuclide, F , similar to Equation (2.11). For example, the total delayed neutron yield is given as:

$$\bar{v}_d = \left(\sum_F \int \Sigma_f^{(F)} \phi dE d\vec{r} \right)^{-1} \sum_F \left(\bar{v}_d^{(F)} \int \Sigma_f^{(F)} \phi dE d\vec{r} \right), \quad (3.11)$$

where,

$$\bar{\nu}_d^{(F)} = \text{the total delayed neutron yield for nuclide } F \text{ [-]} \quad (3.12)$$

$$\Sigma_f^{(F)} = \text{the macroscopic fission cross section for nuclide } F \text{ [cm}^{-1}\text{]}. \quad (3.13)$$

These weighted group parameters are generated using the static and the improved values. Then, they can be used in simple step reactivity insertion transients, where the following four sets of data are compared: static group parameters without diversion, static group parameters with diversion, improved group parameters without diversion, and improved group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and improved group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating DNP group parameters.

Chapter 4

Preliminary Results

4.1 Sensitivity Studies

These sensitivity studies are used to analyze parameters that are held constant during the simulations and determine if simplifications can be made to some with negligible impact on the results. For these simulations, the parameters, unless otherwise specified, are given in Table 4.1.

Parameter	Value	Units
OpenMC Batches	10	-
OpenMC Particles	50000	-
OpenMC Depletion Chain	ENDF/B-VIII.0	-
OpenMC Cross Sections	ENDF/B-VIII.0	-
OpenMC Source Particles	4.2×10^{16}	-
Temperature	920	<i>K</i>
Neutron Energy	0.0253	<i>eV</i>
Sample Nuclide	^{235}U	-
Sample Density	10	$\frac{\text{g}}{\text{cm}^3}$
In-core Residence Time	20	<i>s</i>
Ex-core Residence Time	0	<i>s</i>
Irradiation Time	420	<i>s</i>
Decay Time	420	<i>s</i>
Decay Time Step	1	<i>s</i>

Table 4.1: Default parameters used for sensitivity studies of constant parameters used in 0D flow DNP group parameter formulation OpenMC simulations.

OpenMC batches and OpenMC particles both refer to the settings in OpenMC which dictate the number of batches per generation and the number of particles per batch, respectively. The source particles parameter dictates the strength of the neutron source. The number of source particles is divided by the irradiation time to generate the neutron source rate used in OpenMC during the irradiation, as shown in Equation (3.7). The irradiation time is the amount of time the sample is irradiated, the decay time is the amount of time the sample has the delayed neutron count rate measured after irradiation, and the decay time step is the time step used when updating the nuclide

concentrations post-irradiation. Additionally, by default, no chemical removal is modeled.

4.1.1 OpenMC Particles

If this value can be reduced, then the total amount of simulation time can be reduced. Alternatively, the default value may be insufficient for this problem and lead to inaccurate results. The model is exceedingly simple, composed of only a small sphere of fissile material and a vacuum boundary condition. Table 4.2 shows the total delayed neutron yields and average half-life for varying numbers of simulated particles in OpenMC. Because more particles lead to a slower simulation, a balance must be found between accuracy and cost. The differences between the largest number of particles for the total delayed neutron yield, $\bar{v}_d$, and the average half-life, $\bar{T}$, are given in the final two columns of the table. Based on these values, 50000 particles results in a fit accurate for the average half-life and total yield while also being somewhat computationally inexpensive.

Particles	$\bar{v}_d$	$\bar{T}$ [s]	$ \Delta \bar{v}_d $	$ \Delta \bar{T} $ [s]
1	0.016339	7.343276	0.002394	0.028612
10	0.019469	7.313167	0.000736	0.001497
100	0.018841	7.314828	0.000108	0.000164
500	0.018810	7.314185	0.000077	0.000478
1000	0.018693	7.315162	0.000040	0.000499
5000	0.018667	7.314440	0.000066	0.000223
10000	0.018700	7.314947	0.000033	0.000284
50000	0.018741	7.314586	0.000008	0.000077
100000	0.018730	7.314608	0.000002	0.000055
500000	0.018731	7.314636	0.000002	0.000028
1000000	0.018733	7.314663	-	-

Table 4.2: Sensitivity study of total delayed neutron yields and average half-lives due to number of particles simulated in OpenMC.

The variation in these values for lower particle counts is driven by differences in the group 2, 3, and 4 fits. Figure 4.1 shows the group yields for each different number of particles, where the largest differences are in groups 2, 3, and 4. These group half-lives range from 4.4 to 24 seconds. The cause of this difference must be from differences in nuclide concentrations post-irradiation, as the emission probability and decay constant data are the same between the different simulations compared here. To evaluate which nuclides are causing this difference, the standard deviation of each set of concentrations is divided by the mean, which results in the coefficient of variation (CV) for each nuclide. Some of the nuclides with the largest CV have a negligible contribution to the delayed neutron yield, such as ^{209}Tl . To scale by this consideration, the CV is multiplied by the total number of delayed neutrons emitted from that nuclide i , $\bar{v}_{d,i}$, the equation for which is given in Equation (2.22), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}$$

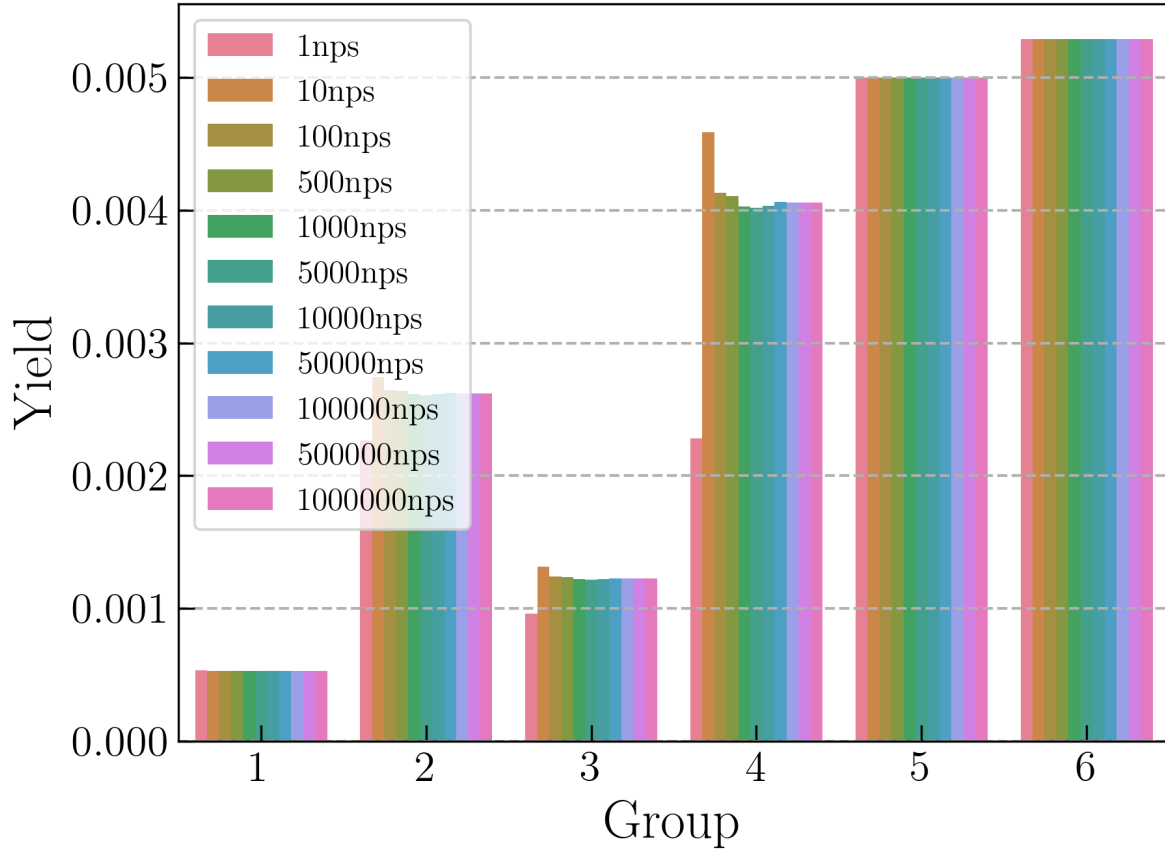


Figure 4.1: Group yields resulting from various numbers of simulated particles in OpenMC.

The scaling of the CV by the total delayed neutron yield is important because nuclides like ^{209}Tl have a CV of 9.65%, but a total delayed neutron yield of 0. The nuclides are sorted by a new scoring term, α , where the full term in Equation (4.1) is given as:

$$\alpha_i = \frac{\bar{v}_{d,i} CV}{\bar{v}_d} \quad (4.1)$$

Based on α , only two of the top ten nuclides are within groups two through four, with the nuclides being ^{89}Br and ^{137}I . When sorting by CV instead, none of the top thirty nuclides are within groups two through four. If the CV is replaced with the standard deviation, then there are six out of ten nuclides which are within groups two through four. These nuclides are ^{137}I , ^{88}Br , ^{93}Rb , ^{89}Br , ^{138}I , and ^{136}Te . Although the relative changes in these nuclides are smaller, the absolute change is larger, which drives a greater difference in groups 2 through 4.

4.1.2 Temperature

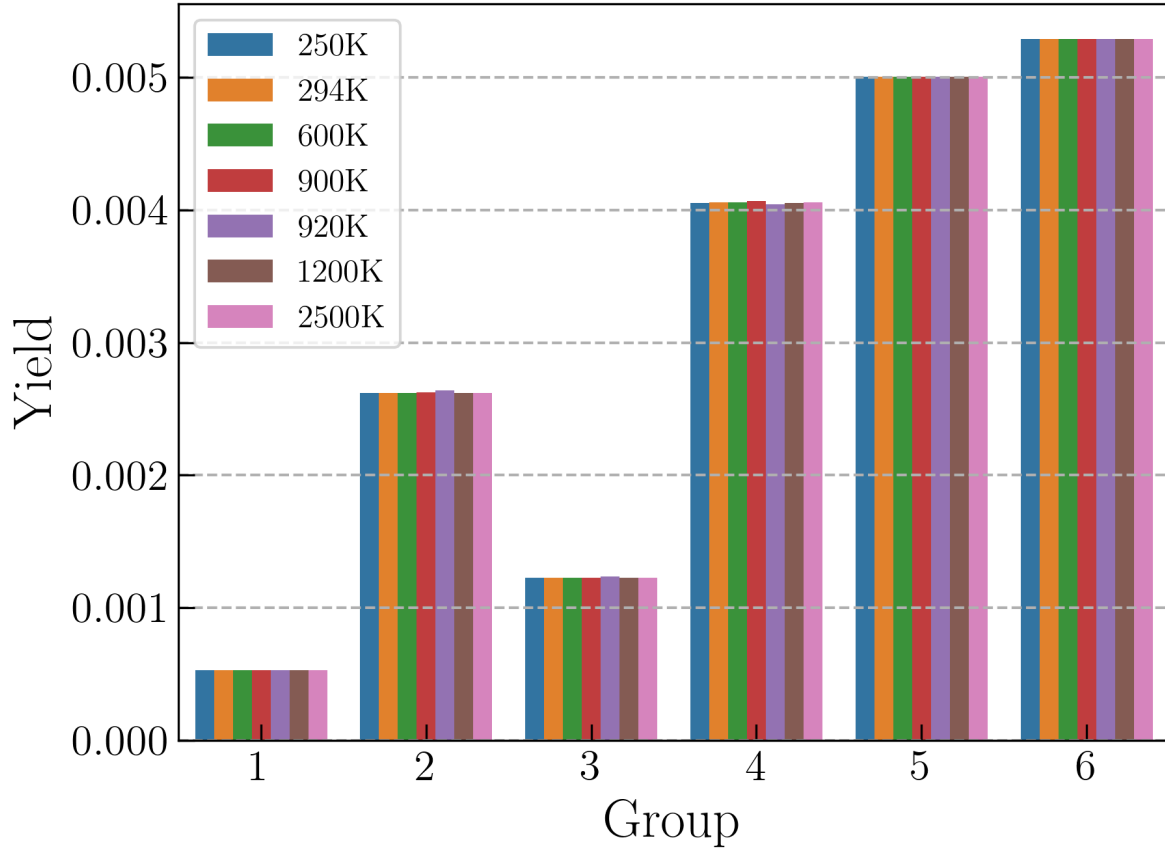


Figure 4.2: Group yields resulting from various sample temperatures in OpenMC.

Temperature [K]	$\bar{\nu}_d$	$\bar{T}$ [s]
250	0.018729	7.315157
294	0.018732	7.314929
600	0.018735	7.314949
900	0.018743	7.314833
920	0.018745	7.318251
1200	0.018722	7.314862
2500	0.018732	7.315089

Table 4.3: Sensitivity study of total delayed neutron yields and average half-lives due to the sample temperature simulated in OpenMC.

4.1.3 Target Density

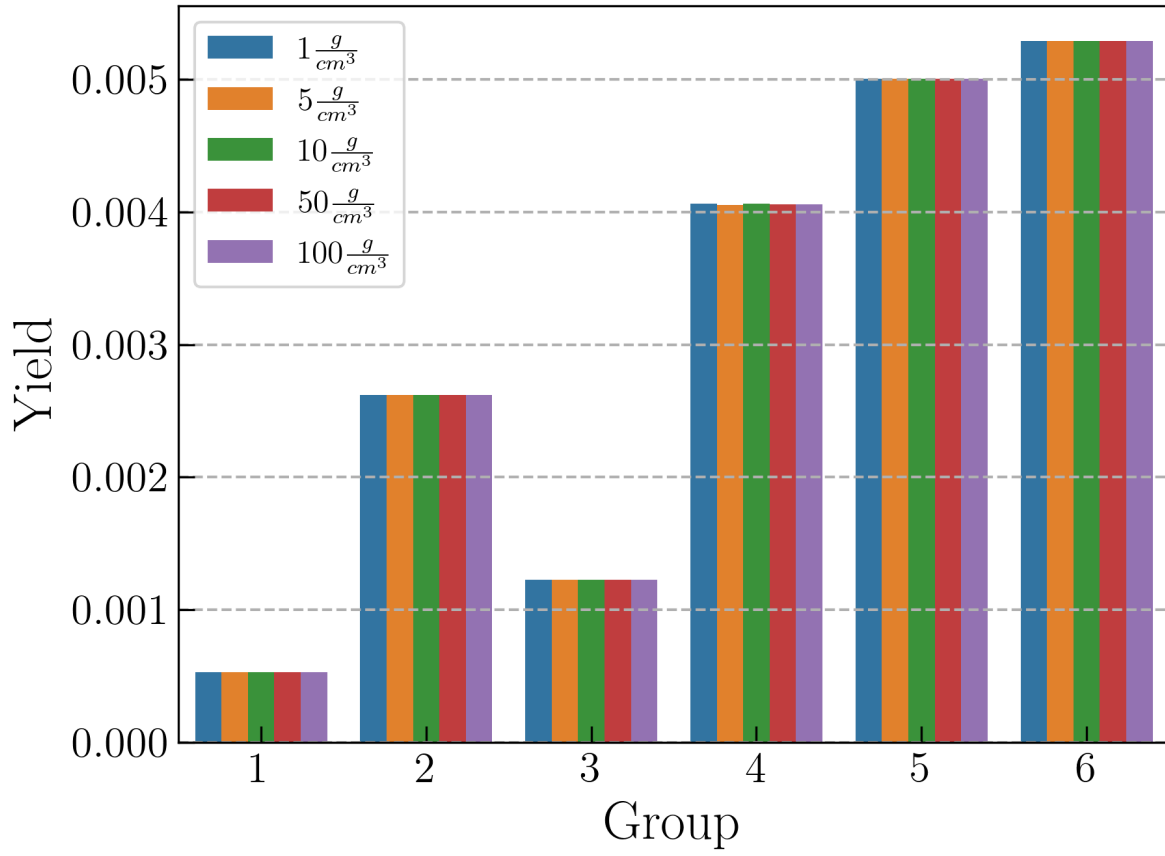


Figure 4.3: Group yields resulting from various sample densities in OpenMC.

Density [$\frac{g}{cm^3}$]	$\bar{\nu}_d$	$\bar{T}$ [s]
1	0.018738	7.313331
5	0.018725	7.313464
10	0.018735	7.313371
50	0.018731	7.313418
100	0.018730	7.313413

Table 4.4: Sensitivity study of total delayed neutron yields and average half-lives due to sample density simulated in OpenMC.

4.1.4 Irradiation Time

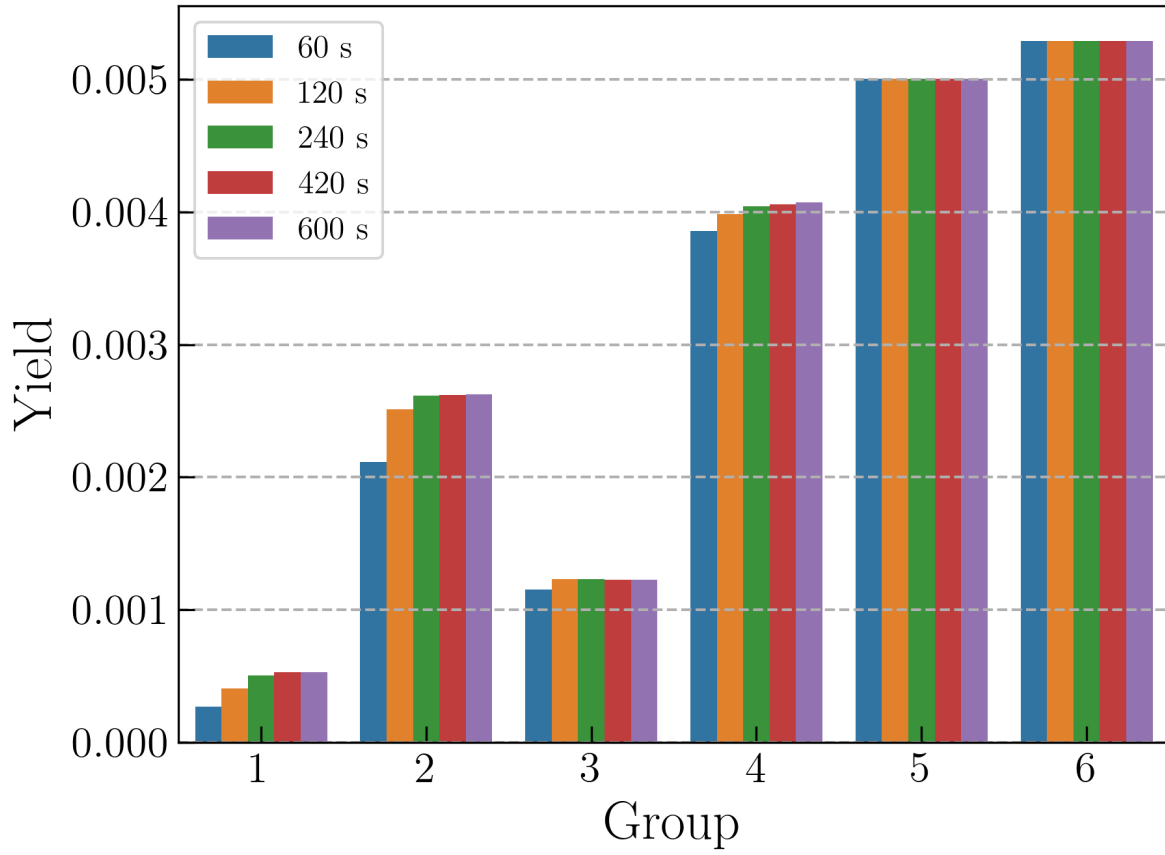


Figure 4.4: Group yields resulting from various irradiation times in OpenMC.

Time [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
60	0.017690	7.388026	0.001062	0.066482
120	0.018433	7.355979	0.000319	0.034434
240	0.018687	7.329571	0.000065	0.008026
420	0.018731	7.322325	0.000021	0.000780
600	0.018752	7.321545	-	-

Table 4.5: Sensitivity study of total delayed neutron yields and average half-lives due to net irradiation time simulated in OpenMC.

4.1.5 Decay Time and Decay Time Steps

The bulk of the computational cost comes from writing statepoint files; sufficient resolution is required to ensure a smooth decay curve.

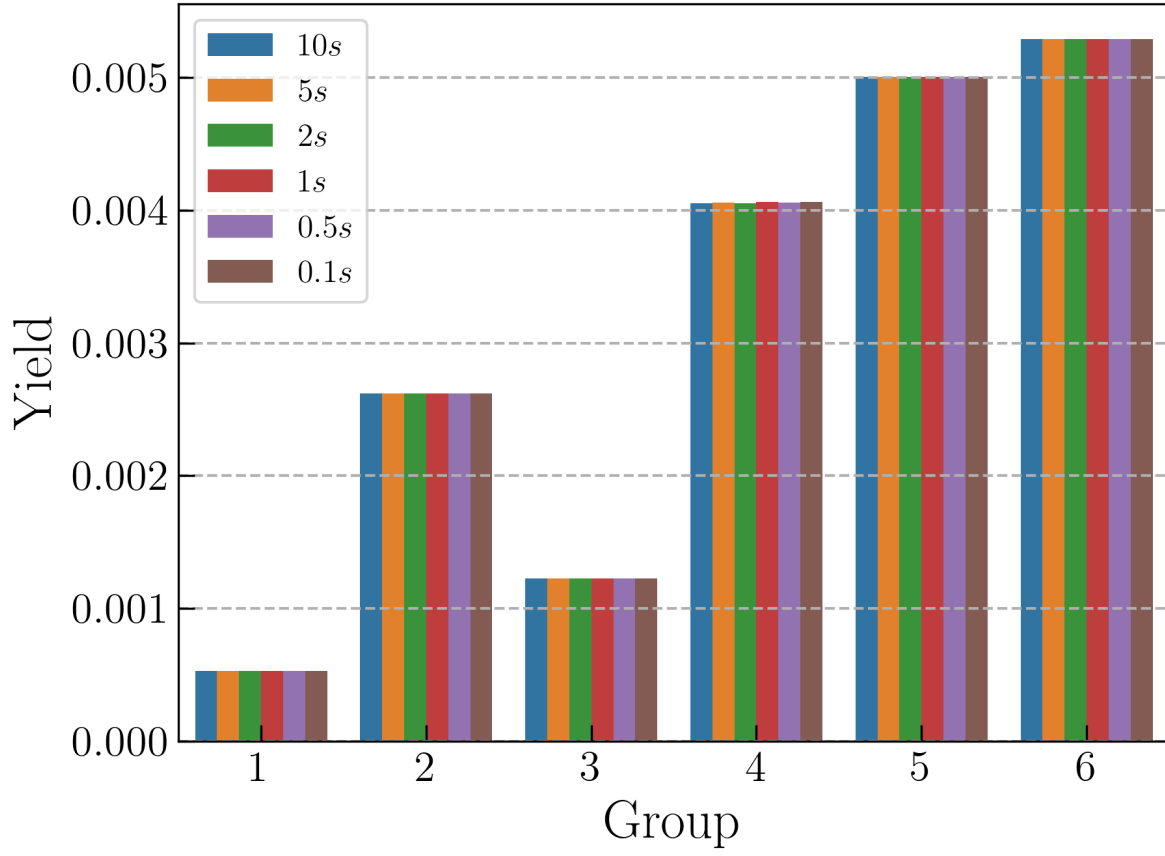


Figure 4.5: Group yields resulting from various decay time steps in OpenMC.

Δt [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
10s	0.018722	7.314094	0.000013	0.000076
5s	0.018732	7.314083	0.000003	0.000065
2s	0.018729	7.314162	0.000006	0.000144
1s	0.018735	7.314021	0.000000	0.000003
0.5s	0.018731	7.314008	0.000004	0.000010
0.1s	0.018735	7.314018	0.000000	0.000000

Table 4.6: Sensitivity study of total delayed neutron yields and average half-lives due to decay time step simulated in OpenMC.

Using a time step of 1 second:

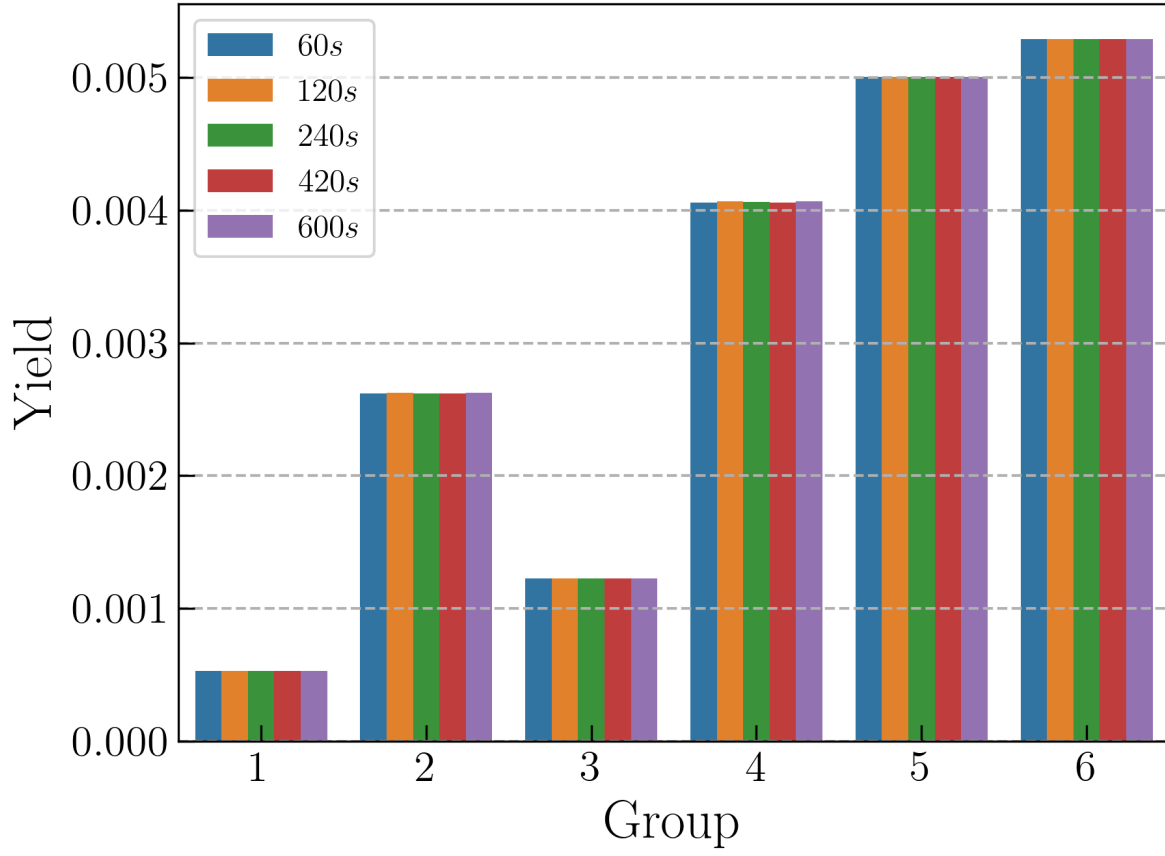


Figure 4.6: Group yields resulting from various decay times in OpenMC.

t [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
60s	0.018734	7.315858	0.000010	0.000039
120s	0.018744	7.315819	0.000000	0.000000
240s	0.018735	7.315828	0.000009	0.000009
420s	0.018734	7.315880	0.000010	0.000061
600s	0.018744	7.315819	0.000000	0.000000

Table 4.7: Sensitivity study of total delayed neutron yields and average half-lives due to decay time simulated in OpenMC.

4.1.6 Large Cycle Time Elements

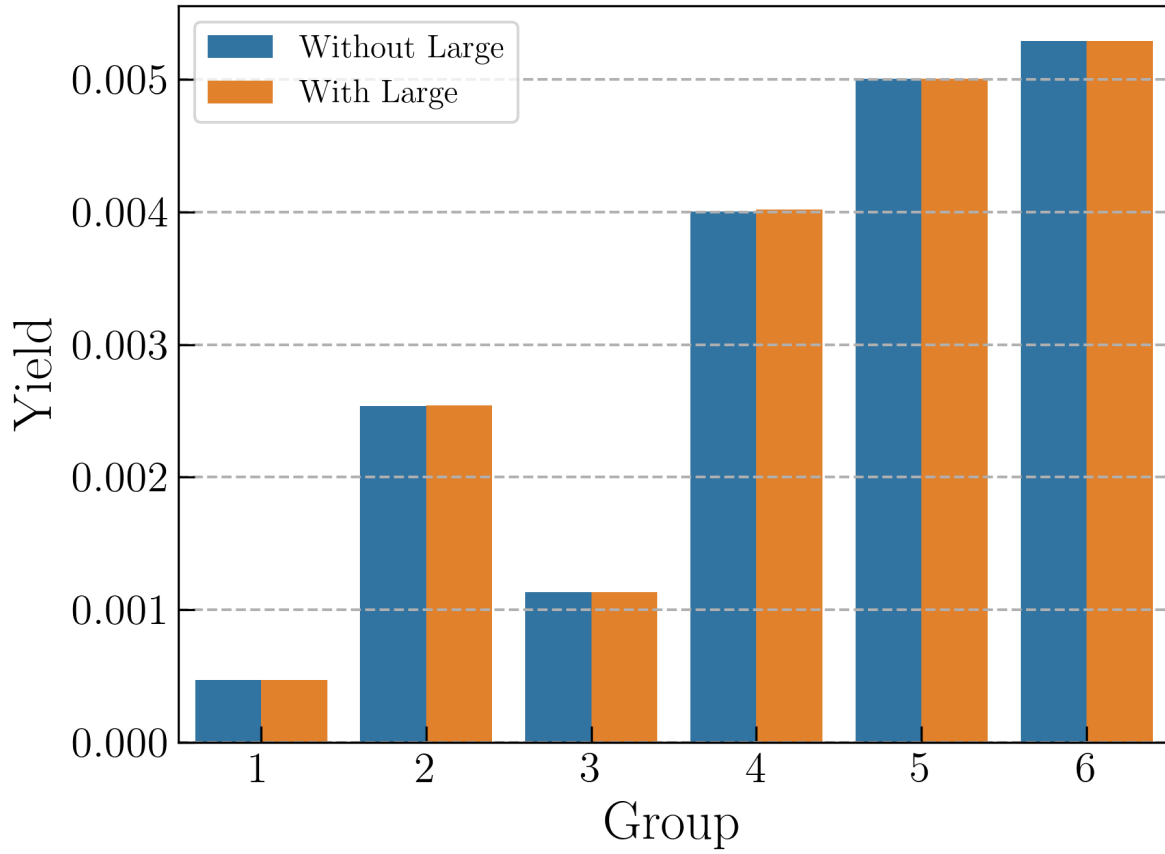


Figure 4.7: Group yields with and without large cycle time reprocessing in OpenMC.

Large Cycle Time Reprocessing	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
False	0.018442	7.060040	0.000018	0.000029
True	0.018460	7.060011	-	-

Table 4.8: Sensitivity study of total delayed neutron yields and average half-lives due with and without large cycle time reprocessing simulated in OpenMC.

4.1.7 Decay Daughter Tracking

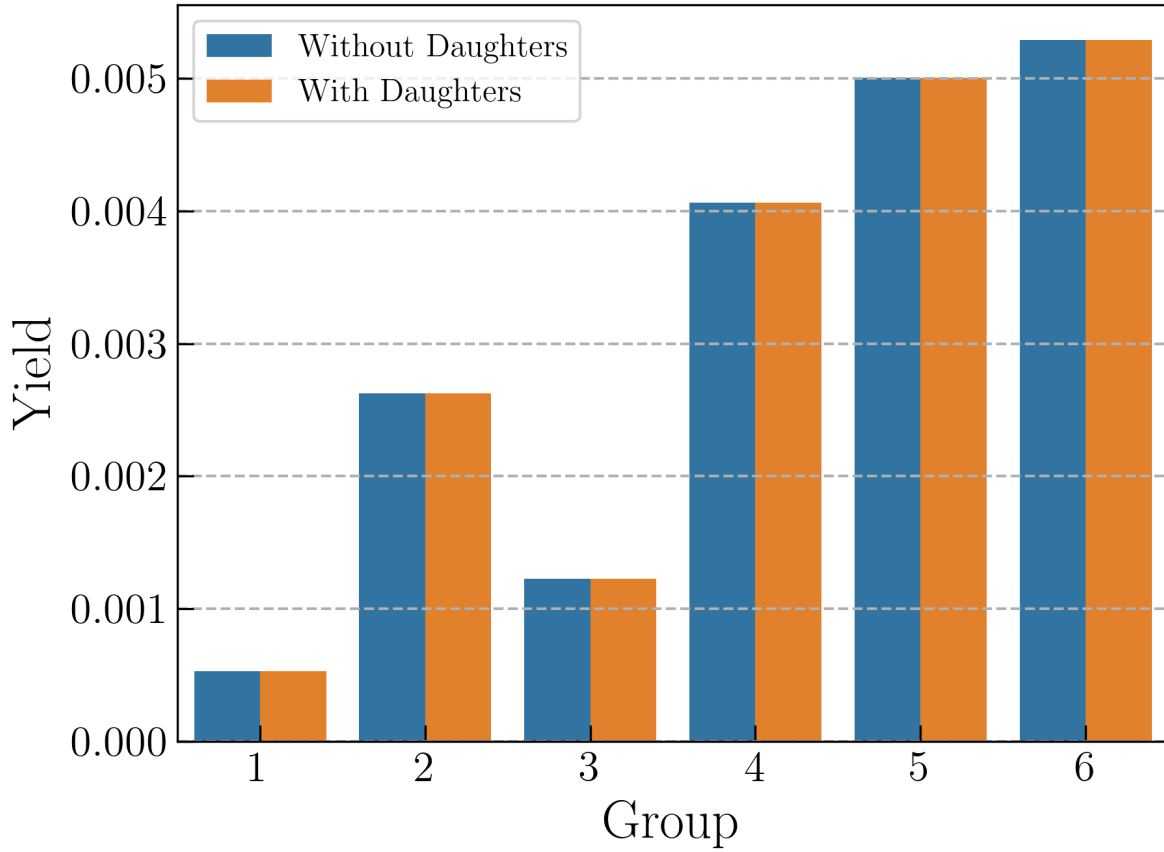


Figure 4.8: Group yields with and without decay chains simulated after irradiation in OpenMC.

Decay Daughters Modeled	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta\bar{\nu}_d $	$ \Delta\bar{T} $ [s]
False	0.018740	7.315761	0.000001	0.000109
True	0.018741	7.315652	0.000000	0.000000

Table 4.9: Sensitivity study of total delayed neutron yields and average half-lives due with and without decay chains simulated after irradiation in OpenMC.

4.1.8 Takeaways

50k particles (0.8 pcm difference very small)

Decay time step of 1 second (becomes more expensive for more time steps) Irradiation time of 600 seconds (difference of 2.1 pcm small, but might as well account for it) Large cycle time elements have 1.8 pcm difference, but negligible computational cost.

All these small differences add up. Do a comparison between the full fidelity results and results with +/- 2 pcm.

Decay daughters don't need to be tracked, but cost is small (many depletion simulations, but they proceed rapidly).

4.2 Verification of 0D Flow Model

[100] table 2 has group parameters, these can be used as a direct comparison and the effects can be compared for the MSFR.

[112] has MSFR PRK and IPK which can be compared.

fluid flow velocity change on power (Dulla PhD thesis I can't access)

[94] - can compare reactivity insertion effects on power

[112] - extracted power effect on power

[109] -

Microscopic group parameters [95, 115, 76] should match static results (no flow or chemical removal.)

ENDF Table II in [76] shows a yield of 1.9 for thermal u235 in ENDF, which matches the static results in the simplified model simulation.

4.3 Effects of MSR Phenomena on Group Parameters

These sensitivity studies are used to analyze the effects of the parameters that represent MSR physical phenomena. Polynomial fits of group parameters. Using MSRE temperature and salt density [144]. No reprocessing unless specified.

The in-core residence time is defined as a minimum, as based on the net residence times, the cycle may not complete. For example, Figure 4.9 shows the irradiation of a sample for various combinations of in-core and ex-core residence times. The net irradiation time is 420 seconds, for which 10, 20, and 30 are all multiples. However, 40 is not, which leads to a situation where the concentrations from the 20 second in-core and ex-core irradiation are from the incorrect cycle. To account for this, the irradiation time is instead treated as a minimum irradiation time, and the actual irradiation time will be increased to allot for the full cycle to complete.

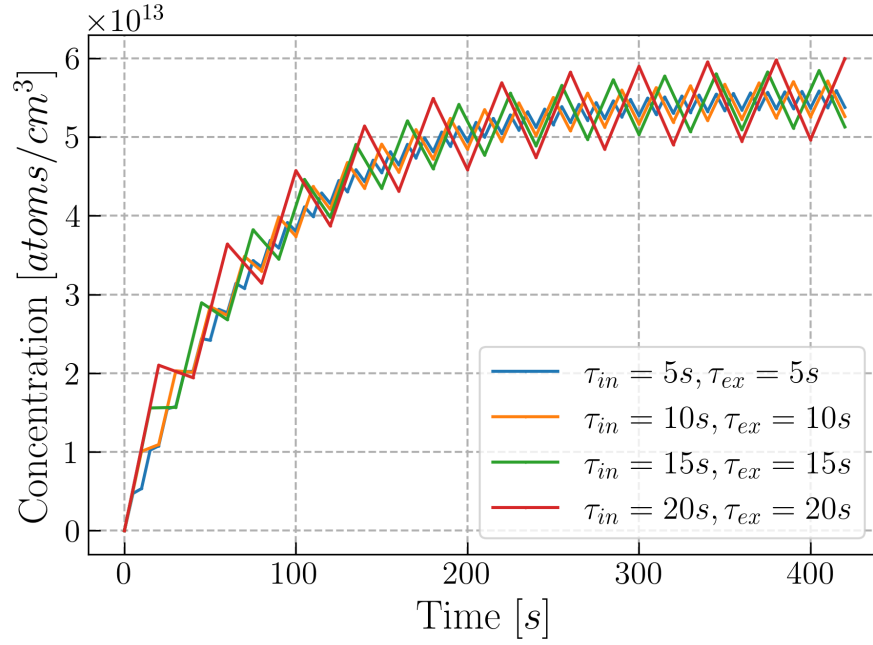


Figure 4.9: A comparison of various in-core and ex-core residence times on the concentration of ^{87}Br during irradiation of the sample over 420 seconds.

Parameter	Value	Units
OpenMC Batches	10	-
OpenMC Particles	50000	-
OpenMC Depletion Chain	ENDF/B-VIII.0	-
OpenMC Cross Sections	ENDF/B-VIII.0	-
OpenMC Source Particles	4.2×10^{16}	-
Temperature	920	K
Neutron Energy	0.0253	eV
Sample Nuclide	^{235}U	-
Sample Density	2.3275	$\frac{g}{cm^3}$
In-core Residence Time	-	s
Ex-core Residence Time	-	s
Minimum Irradiation Time	600	s
Decay Time	600	s
Decay Time Step	1	s

Table 4.10: Default parameters used for study on MSR phenomena in 0D flow DNP group parameter formulation OpenMC simulations.

Removal Rates

In and Ex-core Residence Times

Before studying the effects of varying the in-core and ex-core residence times, it is useful to first study a test case with a predictable result. Specifically, a static problem is studied, where only the in-core residence time varies while the ex-core residence time is 0 seconds. The predicted result is that adjusting the in-core residence time only affects the frequency of transport simulations in OpenMC, which should have a minimal impact on the DNP group parameters. This is shown in Figure 4.10, where the results generally agree, with total delayed neutron yields within 2 *pcm*. However, the 25 second in-core residence time has group yields which are noticeably smaller, resulting in a total delayed neutron yield approximately 10 *pcm* smaller than the other fits.

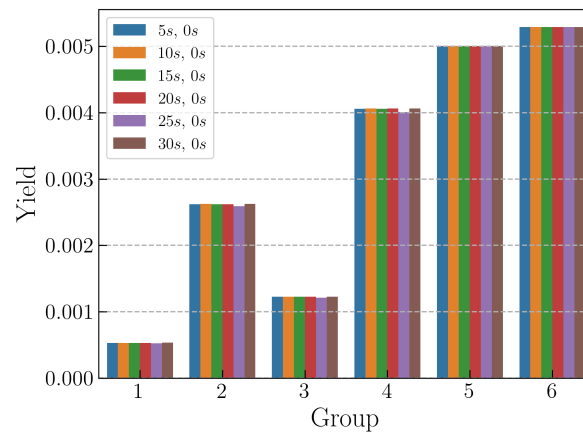


Figure 4.10: Group yields for a static irradiation in OpenMC with varying in-core residence times.

MSRE is approximately 10 seconds in-core and 15 seconds ex-core [144].

explain why the 25 second fits are smaller.

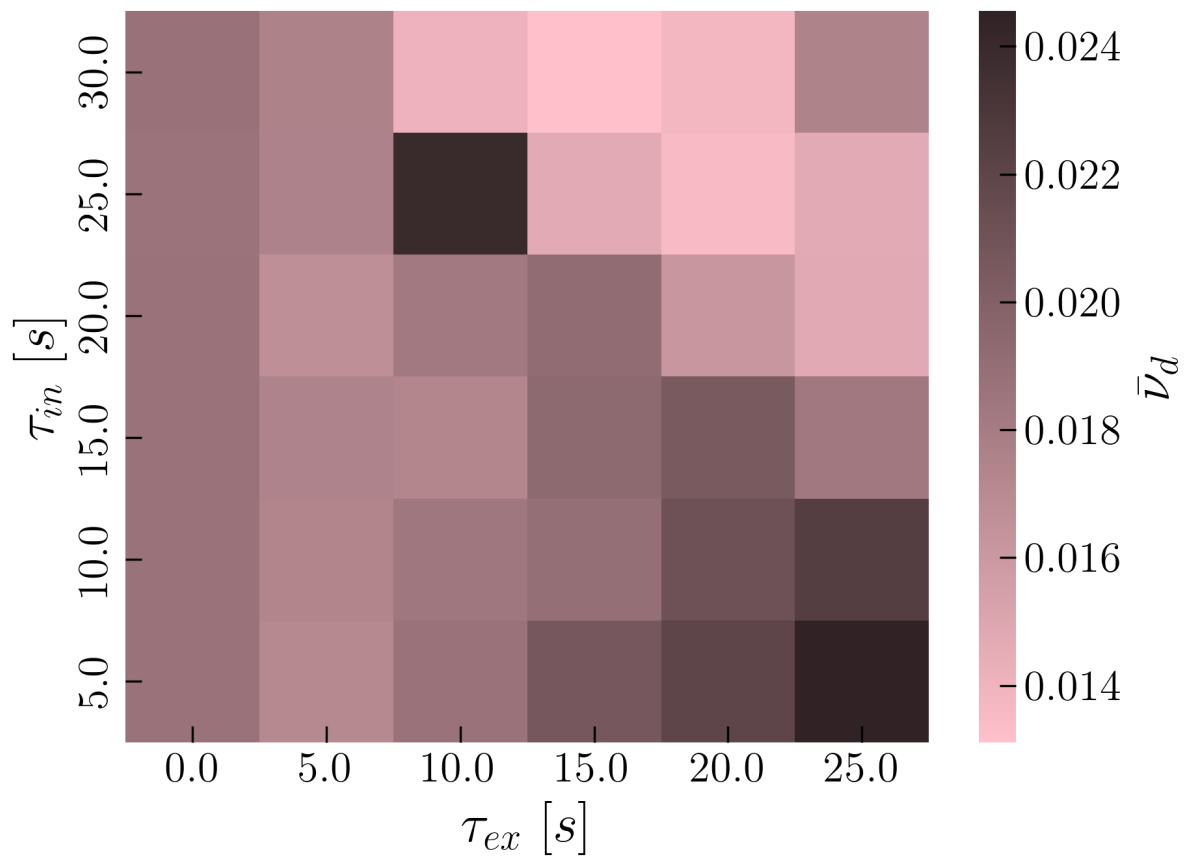


Figure 4.11: Total delayed neutron yield for various residence times during irradiation in OpenMC.

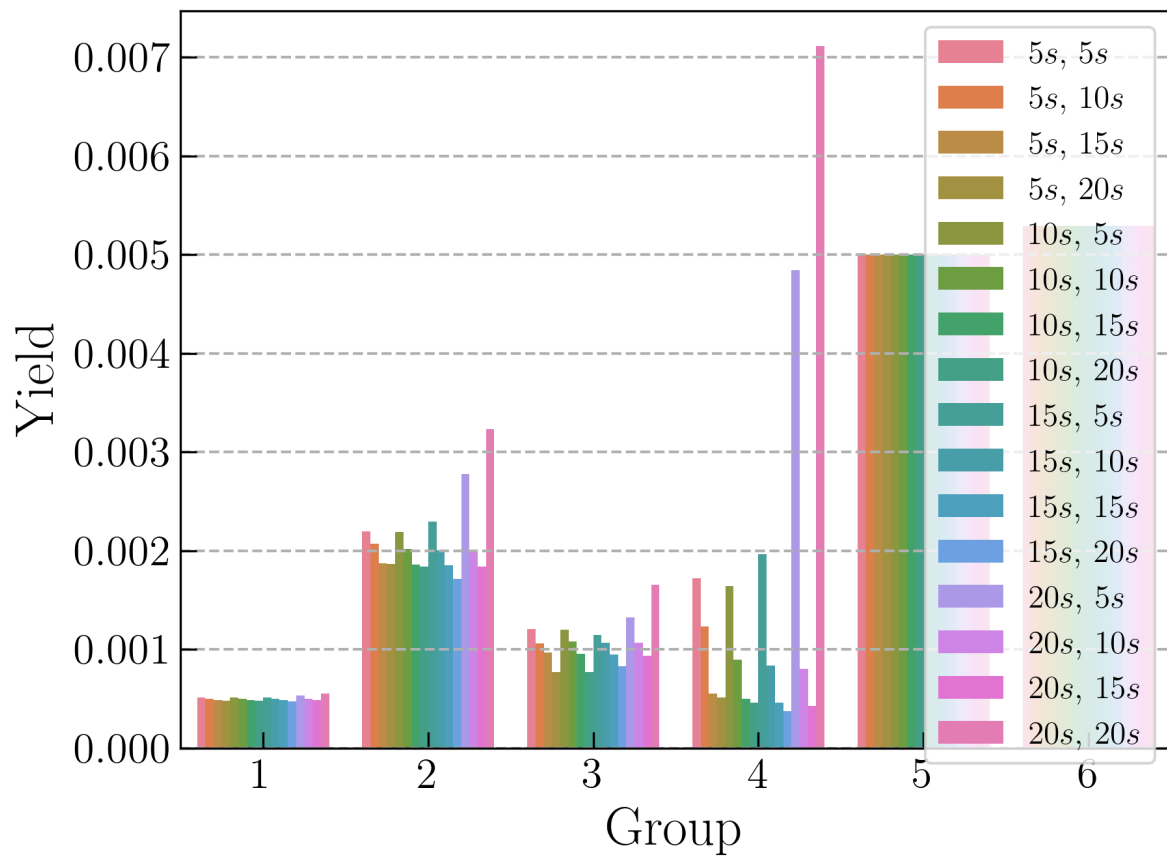


Figure 4.12: Group yields for various residence times during irradiation in OpenMC.

τ_{in} [s]	τ_{ex} [s]	$\bar{\nu}_d$	$\bar{T}$ [s]
5s	5s	0.015940	8.101997
5s	10s	0.015165	8.030291
5s	15s	0.014179	8.293764
5s	20s	0.013929	7.770002
10s	5s	0.015851	8.113214
10s	10s	0.014790	8.234031
10s	15s	0.014109	8.304067
10s	20s	0.013845	7.819340
15s	5s	0.016220	7.869896
15s	10s	0.014718	8.240623
15s	15s	0.014051	8.310107
15s	20s	0.013693	8.191821
20s	5s	0.019775	7.537696
20s	10s	0.014672	8.244369
20s	15s	0.013996	8.313858
20s	20s	0.022849	7.536158

Table 4.11: Effect of residence times on the total delayed neutron yield and average half-life.

Chapter 5

Proposed Work

5.1 Group Parameter Evaluation

It is vital to evaluate the DNP group parameters to ensure they are accurate and usable. To assess accuracy, the parameters are validated using macroscopic experimental data and MSRE transients. To evaluate usability, modified versions of the MSRE and MSFR are analyzed using the spatially resolved flow model. These evaluations demonstrate the validity and practical applications of the DNP group parameters.

5.1.1 Validation

Validating the DNP group parameters is essential to ensure their practical utility. Although the group parameters are verified with other group parameters in the literature, this does not necessarily mean they capture all physical phenomena of interest. Validation identifies potential weaknesses in the approach by highlighting discrepancies between predicted and experimental results.

Validating the DNP group parameters generated in this work is conducted in two stages, as discussed in Section 3.5. The first stage validates the delayed neutron count rate formulation and group generation for a static problem. The second stage validates the improved DNP group parameters by comparing their reactivity feedback with that of the static group parameters.

The first stage validates the irradiation by comparing the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [84]. This requires modeling Godiva in OpenMC and solving the k-eigenvalue problem to irradiate the sample. The post-irradiation sample concentration in MoSDeN is used to generate the DNP group parameters. The delayed neutron count rate and group parameters are compared with values from Keepin et al. to identify discrepancies. Some differences are expected due to uncertainties in the data and experimental results, so validation is successful if the results align within a few percent. If they do not align, this indicates either an issue with MoSDeN or with the data. If the issue lies with the data, it will be noted for future work to improve the microscopic data, enabling a more successful implementation of the proposed methodology.

The second stage validates the improved DNP group parameters. This stage uses MSRE pump start-up and coast-down experimental data to assess the accuracy of the group parameters in modeling reactivity changes from DNPs [136]. This stage requires modeling the MSRE core and upper and lower plena in Moltres [58]. The reactivity effects of the DNPs are calculated for both transients using static and improved DNP group parameters. The improved group parameters are expected to predict reactivity more accurately as they account for more physical phenomena in the MSRE. If neither set of reactivity responses aligns with the experimental results, then there

is a model issue, as Moltres has previously demonstrated the ability to model these transients [139, 125]. These transients have also been evaluated using other tools, providing additional comparison [54, 44]. If the static group parameters predict reactivity more accurately than the improved group parameters, this suggests an issue with the improved group parameter formulation, particularly if the first validation stage succeeded. The issue must be resolved before proceeding, as the improved DNP group parameters should provide higher accuracy in MSR modeling.

5.1.2 Reactor Generalization

One of the desired outcomes for this work is formulation of DNP group parameters for MSRs which can be applied generally. If the DNP group parameters can be applied generally to different systems with similar residence times, then the amount of computational work required is drastically reduced. However, it is necessary that sufficient evidence is provided demonstrating that the improved parameters are identical for different MSRs.

Specifically, the MSRE and the MSFR will be simulated using the spatially resolved model [144, 27]. The models will be adjusted such that the in-core and ex-core residence times are identical for each. The result will be a set of group parameters for two distinct reactors but with the same residence time tabulations. Instead of calculating the neutron flux, both reactors will be given an incident neutron energy spectra for irradiating the sample. The impact of the incident neutron energy spectra is discussed in Section 5.2.4, so differences caused by the spectra will not be studied in this section. Additionally, the fuel salt and fuel temperature in the MSFR will be replaced with those from the MSRE.

The two different reactor simulations will have all of the same input parameters aside from the mesh and thermal-hydraulic parameters. If the DNP group parameters have negligible differences, then the geometry and thermal-hydraulic parameters are not necessary when generating the group parameters. Instead, all that is needed are the other input parameters, meaning the work does not need to be replicated for every different reactor geometry and thermal-hydraulic configuration. If the DNP group parameters have non-negligible differences, then each MSR needs to have DNP group parameters generated for its own specific geometry and flow conditions. Further investigation may also be warranted to determine if there are one or two parameters which could be used to capture these differences, thus reducing the total amount of work necessary to model all MSR designs.

5.2 Data Evaluation

The use and handling of various data directly impact the DNP group parameters. These impacts arise from the data source, time resolution of fission yields, number of DNPs included, and incident neutron energy used to irradiate the sample. The data source includes cross-section data, fission yields, emission probabilities, half-lives, and other associated data used to generate the DNP group parameters. The time resolution of fission yields refers to the difference between using cumulative fission yields and time-resolved depletion chains to evaluate DNP concentrations. The number of DNPs is critical, as modeling more DNPs increases computational cost for the spatially resolved model, so minimizing this term without affecting group generation is desirable. Finally, the incident neutron energy spectra will be investigated to determine the most effective way to organize the DNP group parameters by energy.

5.2.1 Data Sources

Huynh et al. compare the effects of cumulative fission yields and emission probabilities from Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [76]. A similar comparison can also be found in the JEFF report 17 [10]. Mathieu et al. investigate additional datasets, including the National Nuclear Data Center (NNDC) website (which sources data from Evaluated Nuclear Structure Data File (ENSDF)) and various data compilations [109, 131, 7, 150]. These comparisons can be expanded to include the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and other data compilations for emission probabilities (delayed neutron branching ratios), half-lives, fission yields, and cross sections. Comparing these data sources clarifies the uncertainty caused by differences in the data. Table 5.1 shows the data sources used in the comparison. The "Concentration" column refers to data including fission yields and cross-sections that affect the DNP concentration term. Although half-lives also influence concentration, data sources without a concentration contribution provide only half-life and emission probability data. Audi et al., Pfeiffer and Kratz, and Rudstam et al. present data from the literature rather than official databases. Audi et al. use ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz present experimental data, an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. present experimental data.

Data name	Year	Source	Half-life	Emission probability	Concentration
IAEA database	2018	[47]	Yes	Yes	No
JEFF 3.3	2017	[134]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[28]	Yes	Yes	Yes
JENDL-5	2021	[79]	Yes	Yes	Yes
ENSDF	2025	[52]	Yes	Yes	No
Audi et al.	2003	[7]	Yes	Yes	No
Pfeiffer and Kratz	2002	[131]	Yes	Yes	No
Rudstam et al.	1993	[150]	Yes	Yes	No

Table 5.1: Various data sources as well as the data which they contain.

The expected outcome is that recently updated data and official databases will be more accurate than collections from the literature. Recent data should be more accurate due to improved equipment and measurement techniques. Official databases are expected to be more reliable, as they have many users reporting errors and have undergone extensive development over time. Investigating these data sources and comparing the DNP group parameters generated from each will be useful when analyzing uncertainty. If the differences are large, understanding the uncertainty propagation will be important. If the differences are negligible, this will simplify further uncertainty analysis.

5.2.2 Fission Yield Time Resolution

The time resolution for formation of DNPs is irrelevant for stationary, saturation irradiation DNP group parameter formulations, allowing the use of cumulative fission yields instead of full depletion simulations. However, the time resolution of DNP formation needs to be considered for group parameter calculations including circulation. This is an important consideration, as if using the cumulative fission yield generates the same group parameters as the full depletion simulation, then the computational cost would be greatly reduced. It is also useful because the spatially

resolved method using the cumulative fission yield is significantly cheaper than full spatially resolved depletion.

Before comparing the methods for a circulating system, the methods are first compared using a stationary system. The DNP group parameters should align for both methods in a stationary system. This approach is common in the literature, so if the results do not align, this indicates an issue with the implementation.

To evaluate the efficacy of using cumulative fission yields compared to depletion, a direct comparison will be made between the group parameters generated by each method. The same dataset, ENDF/B-VIII.0, will be used for both to eliminate differences from data. The DNP group parameters will be generated using each method for various residence times to capture time-dependent effects at higher resolution.

If the DNP group parameters are identical for all residence times, this will significantly reduce computational costs, and the simpler method will be implemented. If non-negligible differences exist, the more costly full depletion method will continue to be used.

5.2.3 Number of DNP

The number of DNPs to track is of particular interest for the spatially resolved flow model. The 0D flow model's computational cost increases negligibly with more DNPs, while the spatially resolved flow model requires significantly more memory and operations to track each DNP over space and time. To improve computational efficiency, minimizing the number of DNPs is desirable. If the 0D flow model generates DNP group parameters with negligible differences compared to the spatially resolved flow model, minimizing the number of DNPs becomes less critical. However, it is of interest to optimize the spatially resolved flow model for potential future analyses which could include spatially resolved reprocessing and other effects.

The impact of each DNP is given in Equation (2.22), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}.$$

Dividing $\bar{v}_{d,i}$ by the total delayed neutron yield, $\bar{v}_d$, provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact can be modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters. A full sensitivity study will be conducted to determine a reasonable number of DNPs to include in the spatially resolved flow model.

5.2.4 Incident Neutron Energy Spectra

The incident neutron energy spectrum directly affects the DNP group parameters. However, it would be computationally expensive to generate new DNP group parameters for every unique incident neutron energy spectrum. Some coarser resolution is required, so it is of interest to determine a reasonable categorization of the DNP group parameters for different energy spectra.

This sensitivity study will investigate different levels of fidelity for organizing the energy spectra: fine, middling, coarse, and single-value spectra. The approach uses the 0D flow model, with changes only to the neutron source energy spectra. The fine, middling, and coarse spectra include the full incident neutron energy spectra but with different bin sizes. The single-value spectra include several categories: neutron energy classification, mean energy, and median energy. The neutron energy classification is provided in Table 5.2, recreated from [106]. The study will also examine hand-crafted spectra where the mean and median remain unchanged, but the spectrum itself varies.

Category	Energy Range
Cold	$< 0.003 [eV]$
Thermal	$0.003 - 0.4 [eV]$
Epithermal	$0.4 - 100 [eV]$
Intermediate	$0.1 - 200 [keV]$
Fast	$0.2 - 10 [MeV]$
Relativistic	$> 10 [MeV]$

Table 5.2: Classification of neutron energy categories by energy range from [106].

This sensitivity study will determine the necessary level of detail in the incident neutron energy spectrum for generating DNP group parameters. Organizing by category or mean energy is expected to suffice for most cases. For cases where this approach is inadequate, a more robust method can be selectively applied.

5.2.5 Irradiation Combination

A key aspect of the methodology is combining pulse and saturation irradiation DNP group parameters. The current approach is similar to Brady's approach, combining the group 5 and 6 parameters from the pulse irradiation with the remaining parameters from the saturation irradiation to form the six DNP group parameters [24]. Another approach is the Levenberg-Marquardt method, which combines pulse and saturation irradiation data [99, 108, 114]. This method of fitting DNP group parameters to both datasets simultaneously may yield more accurate results. This can also be accomplished using the trust region reflective (TRF) approach, which allows for bounds to be included for the parameters [25].

The proposed methods to compare are the current method, the Levenberg-Marquardt method, and the TRF method. The new methods minimize the residual of the form:

$$r = \min_x \left\| \frac{A_p x - b_p}{b_p + \epsilon} + \frac{A_s x - b_s}{b_s + \epsilon} \right\|_2^2, \quad (5.1)$$

where,

- r = the residual of the problem [-]
- A_p = the pulse irradiation coefficient matrix [s^{-1}]
- b_p = the pulse irradiation delayed neutron count rate data [$n \cdot s^{-1}$]
- A_s = the saturation irradiation coefficient matrix [s^{-1}]
- b_s = the saturation irradiation delayed neutron count rate data [$n \cdot s^{-1}$]
- x = the solution vector containing group yields [-]
- ϵ = a small value to avoid division by zero
 $= 1 \times 10^{-12}$.

This formulation of the residual removes the need to linearize the count rate, as the residual has the count rate in the denominator. Additionally, this formulation simultaneously treats the pulse and saturation irradiation data, generating improved DNP group parameters which fit both sets of delayed neutron count rates.

The results of this analysis will be used to evaluate which method can provide a solution with the least computa-

tional cost while also generating accurate group parameters. The group parameters will be evaluated by calculating the ℓ^2 -norm of the residual for the least squares problem using the improved DNP group parameters over all times for both pulse and saturation irradiations.

5.3 Uncertainty Quantification

Several sources of uncertainty can be considered, including half-lives, emission probabilities, cross-sections, and fission yields. These uncertainties propagate into the DNP group and kinetics parameters [138]. Uncertainties also arise from MSR physical phenomena, including flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. Section 4.3 discusses the sensitivity of the DNP group parameters to changes in MSR phenomena, demonstrating the potential impact of the uncertainties in these values.

These uncertainties are important because they propagate to the DNP group parameters, which affect the reactor's response to different transients. It is important to understand this for safety, security, and safeguards design, as the design should account for these uncertainties.

5.3.1 Data Uncertainties

Uncertainties in the data affect the DNP group parameters through the delayed neutron count rate. The uncertainty in the delayed neutron count rate includes uncertainties in half-lives, emission probabilities, and concentrations. Half-life and emission probability uncertainties are provided by experimental data. The uncertainty in concentrations depends on the method used to generate them. In general, concentration uncertainty arises from fission yields (cumulative or independent, depending on the approach), cross-sections, half-lives, and the neutron flux.

Most uncertainties will be collected from ENDF/B-VIII.0, while the flux uncertainty for the 0D flow model will come from OpenMC. Since Moltres uses a deterministic method to calculate the neutron flux, the flux uncertainty could be calculated using the STOCHASTIC TOOLS module in MOOSE. The STOCHASTIC TOOLS module allows adjustment of other uncertain parameters to calculate the flux uncertainty due to these factors. However, a simpler approach is to use the flux calculated from OpenMC along with its uncertainty. The uncertainties in the data and flux are then propagated through MoSDeN using the Uncertainties package and applied to various analyses [53].

5.3.2 MSR Uncertainties

In addition to data uncertainties, uncertainties exist in MSR phenomena. These include uncertainties in the in-core residence time, ex-core residence time, and chemical removal rates. Section 4.3 discusses the sensitivity of the DNP group parameters to these terms. It is important to understand the potential uncertainty in these terms, as the residence times and chemical removal rates are not without uncertainty.

The residence time correlates linearly with the flow rate, which is dependent on the temperature. Although the flow rate would also have uncertainty based on variance in the pump, this uncertainty is considered outside the scope of this work. The temperature affects fuel salt density, buoyant force, and viscosity, all influencing the flow rate and residence time. The temperature is influenced by the fission rate, which has uncertainties from the flux and fission cross section. For the 0D flow model, the residence time uncertainties can be conservatively approximated by treating the temperature uncertainty as the difference between the inlet and outlet temperatures for the core. Assuming a constant mass flow rate, the density change due to the large temperature uncertainty provides a velocity uncertainty, which determines the residence time uncertainty. If this conservative approach to temperature uncertainty results in negligible residence time uncertainty, it can be neglected for the spatially resolved

flow model. Otherwise, the spatially resolved flow model can account for this uncertainty using the STOCHASTIC TOOLS module to directly determine residence time uncertainties.

The chemical removal uncertainty is challenging to treat, as there are different chemical processes for removing elements, such as sparging the fuel salt in the core, filtering elements with a nickel filter, and removing elements via liquid-liquid reductive extraction [153]. The removal efficiency of xenon in a gas separation system is given by [127, 153] as:

$$\epsilon = \frac{1 - e^{-K_L a A_C L \left(\frac{1}{Q_{salt}} + \frac{RT}{H} \frac{1}{Q_{He}} \right)}}{1 + \frac{RT}{H} \frac{Q_{salt}}{Q_{He}}}, \quad (5.2)$$

where,

R = the universal gas constant [$L \cdot Pa \cdot mol^{-1} \cdot K^{-1}$]

T = the salt temperature [K]

H = Henry's law constant for solute gas [$Pa \cdot mol^{-1} \cdot L$]

Q_{salt} = the volumetric salt flow rate [$m^3 \cdot s^{-1}$]

Q_{He} = the volumetric helium flow rate [$m^3 \cdot s^{-1}$]

K_L = the liquid phase mass transfer coefficient [$m \cdot s^{-1}$]

a = the gas-liquid interfacial area per unit volume [m^{-1}]

A_C = the contactor cross-section [m^2]

L = the contractor length [m].

The reductive liquid-liquid bismuth extraction distribution coefficient and equilibrium constant depend on the element, temperature, and concentration [116]. In general, an increase in the temperature leads to an increased distribution coefficient and a decreased equilibrium constant. The distribution coefficient represents the ratio of the bismuth phase concentration to the salt phase concentration, and can be thought of as an efficiency term that is multiplied with the mass flow rate to determine the removal rate. However, the distribution coefficient changes with temperature and flow rate differently for each element and fuel salt used.

Propagating the uncertainty in the chemical removal of elements from the fuel salt is challenging, as there are many variables to consider. These include the fuel salt composition, flow rate, redox potential, hydrostatic pressure, interfacial tension, and temperature [161, 176]. Thus, generic equations are not formulated for each possible combination of variables. Instead, only the chemical removal of noble gases and noble metals from the MSRE are investigated [103]. The noble gases are actively pumped out via the off-gas system while noble metals, including molybdenum, technetium, ruthenium, rhodium, palladium, silver, and antimony plate out via reduction by UF_6 . Some elements, specifically selenium, niobium, and tellurium, vary in their behavior. These elements only sometimes plates out based on the redox potential of the salt; a well reduced salt causes them to plate out, otherwise they act as salt seekers [82, 103].

The removal rate, λ_r , of noble gases from the pump bowl via the off-gas system is [81, 103]:

$$\lambda_r = \frac{\dot{V}}{V} \epsilon, \quad (5.3)$$

where,

$\dot{V}$ = the salt flow rate through the gas-liquid separator loop [$m^3 \cdot s^{-1}$]

V = the total volume of salt in the primary loop [m^3]

ϵ = the removal efficiency [-].

For the MSRE, the values are a salt flow rate of 0.003153, a total salt volume of 1.996, and a removal efficiency of 0.026 [144]. For the 0D flow model, this removal occurs while the sample is in the ex-core region and stops while the sample is in the in-core region. This is because the removal occurs in the pump bowl and not inside the core. For the spatially resolved flow model, this removal also occurs in the ex-core region, treated in the same way as the 0D flow model.

The removal rate, λ_r , of noble metals via plate-out is [103]:

$$\lambda_{i,l,r} = \frac{h_{i,l} A_l}{V}, \quad (5.4)$$

where,

$h_{i,l}$ = the mass transfer rate of element i to waste stream l [$m \cdot s^{-1}$]

A_l = the surface area of waste stream l [m^2].

The removal of noble metals does not occur in a single region, unlike the noble gas removal. Instead, there is a distribution of noble metals: 50% are in the hastalloy-N surfaces in the loop, 49% are in the ex-core region, and 1% are in the graphite surfaces in the in-core region [82]. Because 99% of the plate-out occurs in the ex-core region, the removal can be treated as occurring only in the ex-core region. The surface areas and mass transfer rates for the noble metals are given in Table 5.3, recreated from [103].

Component	$h_{i,l} [m \cdot s^{-1}]$	$A_l [m^2]$	$\lambda_{i,l,r} [s^{-1}]$
HX	4.657×10^{-5}	2.93×10^1	6.827×10^{-4}
Fuel loop piping, core and pump volute	1.041×10^{-4}	6.60×10^0	3.441×10^{-4}
Core wall cooling annulus	4.318×10^{-5}	1.43×10^1	3.095×10^{-4}
Core graphite (exposed to salt)	5.334×10^{-6}	1.36×10^2	3.637×10^{-4}
Miscellaneous	-	-	1.695×10^{-4}
Bubbles	4.233×10^{-4}	3.205×10^1	6.7978×10^{-3}
Total	-	-	8.6674×10^{-3}

Table 5.3: Noble metal mass transfer rates, surface areas, and resulting removal rates (recreated from [103] with data from [82]).

5.4 Group Spectra Generation

The delayed neutron energy spectra, χ_d is important for accurate prediction of reactor kinetics [42]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.26), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

Because the DNP group yields and half-lives are affected by the MSR physical phenomena, it follows that the DNP group spectra are also affected.

The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.2. The DNP group spectra will be generated using the half-life grouping and fractional grouping methods. The DNP group spectra will then be compared with the spectra generated from a stationary problem with no chemical removal to determine the effect of MSR phenomena on the spectra.

5.5 Spatially Resolved Model Development

Although the 0D flow model provides useful insights, it does not provide a high level of fidelity for any specific geometry. A spatially resolved model will be developed to evaluate the accuracy of the 0D flow model. This model verifies whether the 0D flow model produces sufficiently accurate DNP group parameters. Section 3.2.3 discusses the spatially resolved model in detail. This spatially resolved model will compare multiple levels of fidelity for generating DNP group parameters: 0D flow model using cumulative fission yields, 0D flow model with depletion, spatially resolved flow model using cumulative fission yields, and spatially resolved flow model with depletion. Each step increases computational cost, so if a simpler method generates DNP group parameters with negligible differences, it will be used preferentially.

The process for comparing the 0D flow model and the spatially resolved model is as follows: the spatially resolved model runs and generates DNP group parameters and residence times, the residence times are used to feed into the 0D flow model, the resulting DNP group parameters are compared between the two models. If the results are within the uncertainty bounds of each other, then this indicates that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach will be necessary to properly generate the MSR DNP group parameters.

5.6 Kinetics Simulations

Kinetics simulations are critical for validating the DNP group parameters using MSRE transients and for investigating transients affecting security, safety, and safeguards design decisions. Section 3.5 discusses the proposed simulations in detail. For safety design, the ULOF and UTOP are investigated for various reactor parameters to assess the impact of the improved DNP group parameters. For security design, sabotage-initiated transients are assumed to build off those studied for safety design. For safeguards design, comparisons are made between diversion scenarios with different PR times and material quantities removed. These diversion scenarios alter the fuel composition, affecting the effective delayed neutron fraction and reactor behavior during transients [185]. The improved DNP group parameters will be used to investigate the SNM removal which could remain undetected while using previous DNP group parameters.

The kinetics simulations will use Moltres and the MSRE to demonstrate how the DNP group parameters affect reactivity and temperature responses in various scenarios. These results can be compared with other analyses from the literature [70, 167]. The results will identify which transients are more sensitive to the improved data and highlight potential concerns. A PRK model can also be used for simple reactivity insertion tests with uncertainty tracking to more straightforwardly demonstrate differences between the DNP group parameters.

Appendix A

Derivation of Delayed Neutron Count Rate in 0D Flow

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [187]:

$$R(t) = \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and,

$$\dot{n}_d(T, t) = \bar{v}_d \int_0^T F(\tau) R(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where,

T = the length of the fission history [s]

t = the decay time from the end of the fission history [s]

$F(\tau)$ = the fission rate history [$\frac{\text{fissions}}{\text{s}}$].

Because the results are normalized to a single delayed neutron, the $\bar{v}_d$ term is divided out.

Although the saturation, or infinite, irradiation derivation makes use of assuming the time is infinite, due to the spatial dependence of this problem, the in-core and ex-core timing must be handled properly. For example, if the fission rate history uses a Fourier decomposition of a square wave, then it would not be possible to extract spatial information from this formulation. Specifically, it is important where in the primary loop the sample initially exists. If the sample is initially in the core inlet, then the final portion of the fission rate history would consist purely of decay. If the sample is initially in the center of the core, then the final portion of the fission rate history would consist of irradiation.

For this derivation, the sample is assumed to exist as a point at the core inlet. This means the full cycle consists of an in-core residence time of fission followed by an ex-core residence time of decay, after which the cycle repeats until the net irradiation time is met. The fission rate history is therefore given as:

$$\begin{aligned} \frac{F(\tau)}{F_0} = & H(\tau) - H(\tau - T) + \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \\ & - H(\tau - j \cdot \tau_{in} - (j - 1) \cdot \tau_{ex}), \end{aligned} \quad (\text{AA.3})$$

where,

H = the Heaviside function [-]

$$J = \left\lceil \frac{T}{\tau_{in} + \tau_{ex}} \right\rceil$$

= the minimum number of cycles to meet the irradiation time, T [-]

F_0 = the magnitude of the constant fission rate [$\frac{fissions}{s}$].

Plugging this into Equation (AA.2) for group k yields:

$$\begin{aligned} \dot{n}_{d,k}(T, t) = & \int_0^T H(\tau) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau, \end{aligned} \quad (AA.4)$$

which can be simplified to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \int_0^T H(\tau) e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \end{aligned} \quad (AA.5)$$

Solving the integrals yields:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \frac{1}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) + 0 \\ & + \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + j \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + (j-1) \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (AA.6)$$

The Heaviside functions in the summations resolve to unity while the cycle is within range of the net irradiation time, and become 0 when the cycle time goes beyond the net irradiation time. The problem then simplifies to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = & \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) \\ & + \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (AA.7)$$

If the net irradiation time is assumed to be infinite, then the delayed neutron count rate becomes:

$$\dot{n}_{d,k}(T, t) = F_0 \bar{v}_{d,k} e^{-\lambda_k t}, \quad (\text{AA.8})$$

which matches the result from Wilson and England [187]. However, as previously mentioned, this infinite approach will not work for the spatially dependent analysis. For example, if an in-core residence time is selected which is relatively small compared to the ex-core residence time, Equation (AA.8) would not vary at all, even though the fission rate history would be significantly different. Instead, the problem takes on the form:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} e^{-\lambda_k t}} = 1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (T - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex})}. \quad (\text{AA.9})$$

By isolating the delayed neutron count rate and summing over all groups, the finalized equation can be formulated as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} - \bar{v}_{d,k} e^{-\lambda_k t} e^{-\lambda_k T} \quad (\text{AA.10})$$

$$+ \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (j \cdot \tau_{in} + (j-1) \cdot \tau_{ex} - T)}, \quad (\text{AA.11})$$

which can be rewritten as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (T - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex})}\right). \quad (\text{AA.12})$$

If the in-core residence time is zero, then the delayed neutron count rate goes to zero, which is the expected behavior.

Appendix B

Derivation of Equilibrium Concentration from Cumulative Fission Yield

This derivation begins with the Bateman equation, given in Equation (2.49) and recreated here as:

$$\frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j},$$

where,

- N_j = the atomic density of isotope j $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i $[\text{cm}^2]$
- ϕ = the spectrum-averaged scalar flux in the fuel region $\left[\frac{\text{neutrons}}{\text{cm}^2 \text{s}} \right]$
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ $[\text{s}^{-1}]$
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[\text{cm}^2]$
- N_i = the atomic density of isotope i $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- λ_j = the decay constant of isotope j $[\text{s}^{-1}]$
- $\lambda_{r,j}$ = the removal constant for removal of isotope j $[\text{s}^{-1}]$
- σ_j = the microscopic total transmutation cross-section of isotope j $[\text{cm}^2]$
- $\dot{N}_{f,j}$ = the feed rate for isotope j $\left[\frac{\text{atoms}}{\text{cm}^3 \cdot \text{s}} \right]$.

This can be rewritten to the simplified form given in Equation (2.52), recreated here as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t),$$

where,

$$\begin{aligned}
 S_j &= \text{the source term for nuclide } j, \\
 &\quad \text{containing fission, decay, and transmutation sources } \left[\frac{\text{atoms}}{\text{cm}^3 \text{s}} \right] \\
 \lambda_{net,j} &= \text{the loss term for nuclide } j, \\
 &\quad \text{containing decay and transmutation losses } [\text{s}^{-1}].
 \end{aligned}$$

This can be rewritten using the scaled flux method, which is useful for flowing fuel reactors as it scales the flux term by the fraction of fuel salt in the core. This changes Equation (2.51) to Equation (BB.1), assuming chemical removal only in the ex-core region, given as:

$$\frac{dN_j}{dt} = f_{in}S_j(t) - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t), \quad (\text{BB.1})$$

where,

$$\begin{aligned}
 f_{in} &= \text{the fraction of fuel salt in the core} \\
 f_{ex} &= \text{the fraction of fuel salt not in the core} \\
 \lambda_{r,j} &= \text{the chemical removal rate of } j.
 \end{aligned}$$

The source term, S_j , can then be written using the cumulative fission yield as shown in Equation (BB.2). The cumulative fission yield is the total number of some nuclide formed per fission event, including decay chains that lead to that nuclide. Because the cumulative fission yield captures decays into the nuclide j , the $\lambda_{i \rightarrow j}$ term is not included. Additionally, parasitic absorption is neglected in this derivation, which leads to:

$$S_j(t) = CFY_j \phi \Sigma_f, \quad (\text{BB.2})$$

where,

$$CFY_i = \text{the cumulative fission yield of nuclide } i.$$

Finally, the fission reaction rate term, $\phi \Sigma_f$, is assumed to be equal to 1. Equation (BB.3) can be used to calculate the concentration as a function of time using cumulative fission yields, given as:

$$\frac{dN_j}{dt} = f_{in}CFY_j - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t) \quad (\text{BB.3})$$

Assuming a stationary fuel with no recirculation or chemical removal, the equation simplifies greatly. The solution to that equation, assuming no initial concentration of nuclide j , is given as:

$$N_j(t) = \frac{CFY_j}{\lambda_j} (1 - e^{-\lambda_j t}) \quad (\text{BB.4})$$

As previously mentioned, the cumulative fission yield gives the total amount of the nuclide generated including from decay chains, so the time dependent result is non-physical. For a saturation irradiation however, the sample is irradiated for a sufficiently long time such that all DNPs reach equilibrium. This means that the equation should be

taken as time approaches infinity, which leads to the form [76]:

$$N_j(t) = \frac{CFY_j}{\lambda_j}. \quad (\text{BB.5})$$

For the case of a nuclide in a circulating fuel reactor with a non-zero chemical removal rate, the concentration as a function of time comes out as:

$$N_j(t) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \left(1 - e^{-f_{ex}\lambda_{r,j}t - \lambda_j t}\right) \quad (\text{BB.6})$$

Equation (BB.6) can also be taken as time goes to infinity, resulting in:

$$N_j(t) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \quad (\text{BB.7})$$

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